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*Simulating the effects of HIV/AIDS disease on human population and
exploring the management option using Agent Based Model and Integrating
Machine Learning for Predictive Disease Outcome Analysis*

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ABSTRACT

This project aims to simulate the effects of HIV/AIDS disease on human population using agent-based model and explore the management options and predict the future disease outcomes by integrating machine learning algorithm with the ABM. The data set containing demographic factors, population density, HIV disease diagnosis and intervention strategies was collected and gathered. The agent-based model was designed to simulate interactions over time and model the transmission of HIV/AIDS among individuals based on the behaviors, demographics and health statuses. Simulation scenarios demonstrated the general spread of HIV, disease progression, deaths caused by the disease, and the effects of interventions, with results visualized using line plots showing the variations in transmission. The simulation data produced by the ABM was subjected to a Random Forest regression approach in order to improve the predicted accuracy of the model. In order to accurately forecast future infection counts, this machine learning model used feature engineering, including lagging to identify temporal relationships in the disease progression. The integration of Random Forest verified its resilience in managing numerous input characteristics and nonlinear correlations, effectively predicting forthcoming transmission patterns. The results underscored the significance of combining simulation-based strategies like ABM with data-driven techniques such as machine learning for a more accurate analysis of HIV/AIDS transmission. This study provides valuable insights for public health decision-makers on optimizing intervention strategies to reduce the spread of HIV/AIDS.

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I certify that the work presented in the dissertation is my own unless referenced.

Signature.....Keerthana Kakkiriken Poothatta

Date.....05/09/2024.....

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CHAPTER 1: BACKGROUND

1.1 INTRODUCTION

The world has witnessed a surge of infectious diseases that poses a great threat to human population. Among the deadlier infectious diseases, the Human Immunodeficiency Virus (HIV) has persisted as a major concern because of its inmedicable nature and the convolution involved in its management and control. HIV is a viral infection that attacks and weakens the immune system specifically the CD4 cells (or T cells). Body fluids like blood, semen, vaginal fluids, anal fluids, and breast milk are the vectors through which the virus transmits. Habitually, unprotected intercourse, sharing needles and birth are the principal avenues of spreading of HIV virus. (History of AIDS - Epidemic, Timeline & HIV | HISTORY, n.d.).The result of cross infection from chimpanzees to humans is believed to originate the human immunodeficiency virus in Western Africa in the 1920s. HIV leads to acquired immunodeficiency syndrome (AIDS), a chronic, potentially fatal illness that increases a patient's susceptibility to opportunistic infections if treatment is not received. (Patel et al., 2021). Within the decades, the pandemic form of HIV-1 has caused 32 million deaths worldwide and approximately 39.9 million people are living with virus according to WHO (HIV data and statistics). Although the therapeutical medical approach of antiretroviral treatment has decreased the toll of deaths by increasing the viral suppression load to reduce the risk of transmission, HIV continues to pose a serious hazard for public health for years to come.

The vulnerability of human populations to virus outbreaks illustrates the need for efficient public health measures to minimize the spread of infectious diseases. Controlling future outbreaks requires an understanding of the mechanism of transmission and the development of practical mitigation technique. With advancements in computer technology, realistic epidemic spread models have been developed by understanding human mobility patterns. Traditional epidemic models such as Compartment models, and complex network models was devised to model the population and disease spread via a dynamical system. (Hackl & Dubernet, 2019) .Although the traditional epidemiological models, such as the SIR (Susceptible-Infectious-Recovered), and types of compartment models such as dynamic and linear models have been instrumental in understanding the disease dynamics, the model fails to effectively model spatial aspects of the spread of an epidemic, the individual contact process, and the effects of individual behaviors, among others. (Hackl & Dubernet, 2019)

(Hackl & Dubernet, 2019) describes a practical technique for researching epidemics known as agent-based models which is feasible to characterise the behaviour of an individual during the course of an epidemic outbreak. Agent based modelling is a computational tool used to stimulate the actions and interactions of autonomous agents (individual or collective entities) in order to understand the system behaviour and outcomes. Agent-based modelling (ABM) offers a more nuanced approach by simulating the actions and interactions of individual agents, which can represent humans, within a defined environment. The agent-based modelling has been a key component in the development of several epidemiological frameworks because of their capacity to generate emergent macro-effects from micro-rules. (Perez & Dragicevic, 2009). It enables the inclusion of a variety of agent characteristics and behaviours, including age, health, social behaviour, and movement patterns. Through the incorporation of social structures, heterogeneous connectivity patterns, and meta-population grouping at the individual level, agent-based modelling of pandemics recreates entire populations and their dynamics. (Khalil et al., 2012).

On the other hand, machine learning (ML) approaches have transformed predictive modelling and data analysis in a variety of fields. ML algorithms such as Artificial Neural Network, Logistic Regression and Random Forest can be used to analyse massive datasets, find trends, and improve intervention techniques in the context of infectious disease modelling. Machine Learning algorithms can be used to predict outbreak scenarios, assess the impact of different interventions, and recommend optimal strategies based on real-time data. (Whitman & Jayaprakash, 2020).

The intersection of Agent Based Models and Machine Learning holds significant assurance for public health. By integrating both techniques, it is possible to create a potential tool for modelling diseases outbreak and exploring the most effective mitigation measures. These methodologies were employed during the Covid 19 pandemic to model the disease spread, evaluate the effectiveness of social distancing measures and inform policy decisions. (James & James, 2021) The use of ML has proven effective for specific disease problems or certain purposes such as disease detection, risk prediction, and spread estimation. (Riswantini & Nugraheni, 2022)

1.2. AIMS AND OBJECTIVES

The primary aim of this dissertation is to develop an agent-based model to stimulate the spread of HIV /AIDS virus among human population and to explore management and mitigation strategies and integrate machine learning technique to predict future disease outcomes. This research aims to enhance our understanding of HIV/AIDS transmission dynamics and optimize intervention strategies such as the load of viral suppression and linkage to care. The objectives of this dissertation are:

Objectives:

- To develop an ABM to model the spread of HIV within a population that accurately simulates individual behaviours and interactions contributing to HIV transmission, based on variables such as gender, age, race, HIV diagnosis, AIDs diagnosis, Viral Suppression and Care linkage data.
- To run simulations under three different scenarios to determine the spread of HIV within the population, across different demographic groups, progression from HIV to AIDS, and to assess the effectiveness of intervention strategies viral suppression and linkage to care on HIV/AIDs transmission. To analyze the significant impact of these factors on the epidemic's trajectory.
- To employ an integrated approach that combines the ABM with machine learning techniques for predictive analysis of future disease outcomes by acquiring the results from agent-based model.
- Based on the simulation and machine learning model results, to offer evidence-based recommendations for public health policies and resource allocation. Highlight the most effective strategies for controlling the HIV epidemic.

1.3 RESEASRCH APPROACH

An approach of quantitative method will be adopted to achieve the objectives this dissertation. A multi methodical approach will be employed by combining the agent-based model and machine learning algorithm.

Literature Review: A comprehensive study of the literature review will be conducted to gather and synthesize on existing research on HIV/AIDS epidemiology models, the use of agent-based modelling in disease spread and the integration of machine learning algorithms. The recent studies will be reviewed and analyzed to identify gaps in current research.

Data Collection and Preparation: Data on epidemiology, population density, demographic factor such as age, gender, race will be collected. Information on the effective interventions such as viral suppression and linkage care will be gathered. Hence the data will be cleaned and processed for the agent-based model.

Developing the agent-based model: An agent-based model will be designed and implemented to stimulate the spread of HIV virus in R platform. Agents representing the individuals with attributes such as age, gender, race and health status will be defined. The probabilities of the disease transmission upon contact between infected and susceptible agent will be defined.

Simulation and scenario analysis: Simulations under three scenarios will be running and key metrics like infection curves depicting infected individuals, number of individuals with AIDs, susceptible individuals, and total number of deaths will be recorded and visualized using plots to analyze the impact of different intervention strategies on the spread of HIV.

Integration of Machine Learning: Machine learning technique will be applied to analyze the simulation outputs and optimizing the intervention strategies. Relevant features such as health status and intervention types from simulation data will be extracted using feature engineering. The simulation data will be prepared by splitting into training and testing set. Random Forest Regression algorithm will be deployed to predict the future disease outcomes. The machine learning model will be trained to using the simulation data to learn the relationship between input features and outcomes.

Evaluation and Validation: Performance of the machine learning model will be assessed by metrics mean absolute error, mean squared error, percentage of variance and visualized using plots. The trained model will be validated using the testing data and compare the predictions against the historical data to assess accuracy.

Documentation: The methodology, results, analysis will be documented in a structured format. The key findings will be illustrated using visual aids such as graphs and charts. Evidence based recommendation will be provided for public health policy and future research.

1.4 OUTLINE OF THE PROJECT

The outline of the dissertation project follows as:

Chapter 2 presents the background work comprising the literature review, critical evaluation and gaps.

Chapter 3 details the methodology of the models developed.

Chapter 4 outlines the results of the models followed by discussion and future work.

Chapter 5 summarizes the conclusion of the project.

CHAPTER 2: LITERATURE REVIEW

The human immunovirus virus (HIV) also known as Acquired deficiency syndrome remains to be a significant challenge to the world. According to UNAID (UNAIDS Global HIV & AIDS statistics) it is currently estimated that 39 million individuals are currently living with virus. HIV infection is the primary cause of morbidity and mortality worldwide with Sub Saharan Africa accounting for majority of the cases. Throughout the 1900s the virus most likely moved sporadically from non-human primates to human. However, it wasn't until homosexual males in urban areas displayed severe and unexplained deficiency that grabbed the attention globally. (Deeks et al., 2015)

HIV is predominantly transmitted through bodily fluids by sexual activity, vertical transmission or inoculation.(Deeks et al., 2015) Following the transmission event, the virus settles in the mucosal tissue, targeting the CD4 cells or T cells and spreads the lymphoid organs within days. The virus first appears in the blood on day 10 and spreads exponentially the following weeks and peaking around day 30 when HIV antibodies are detectable. Individuals are highly contagious at this stage. The immune system then partially controls the virus, establishing a stable 'set point' in HIV replication that can last for years. Profound immunodeficiency is developed in individuals after several years which leads to oncological complication or infectious disease which is defined as AIDS. The prevalence of HIV is higher in specific groups with similar risk factors. Men who have intercourse with men, intravenous drug users, inmates and other closed environments, sex workers, and transgender individuals are among the main groups impacted. (Deeks et al., 2015)Despite the remarkable quantity of research on the virus, no effective treatment strategies which also includes, ART (Antiretroviral therapy) which suppresses the viral infection and improves the immune system, can stop the virus developing into AIDS. Unknown mechanism of the host pathogen is the primary reason for this which necessitates the development of computer technical models to curb the disease. Researchers believe that patients could benefit from the modelling tools with knowledge molecular biology of HIV and advances in diagnostic and therapeutic approaches. Infectious disease mathematical models, simulation models and the use machine learning algorithms are increasingly being used to depict the dynamics of transmission, treatment and prevention which is also recognized in HIV as a potent predictive tool that guide clinical trial design and influence policy.(Hadjiandreou et al., 2007)

The aim of this literature review is to synthesize the existing research to understand the traditional HIV/AIDS epidemiology models, the use of agent-based models in simulating the HIV spread, the use of machine learning algorithms for exploring the mitigation strategies, and finally the integration of agent-based models and machine learning leading to any future works and gaps.

2.1 HIV/AIDS EPIDEMIOLOGY MODELS

In several years it has been made evident by several epidemic outbreaks including HIV, the severe acute respiratory syndrome (SARS) in 2003, the 2009 H1N1 influenza pandemic, the Ebola virus in 2013 and the Zika virus epidemic in 2016, that the disease started locally but spread rapidly around the globe. Consequently, significant research efforts over the past decade have focused on developing quantitative models for understanding the spread of epidemics. Hence numerous models with varying levels of complexity have been devised in the past. A conventional approach for modelling is through a dynamical system, in which the dynamics of the diseased population is given by a system of coupled nonlinear differential equations, is one less intricated model. Conversely, complex network models are more advanced which uses modern network theory methods to represent the underlying social network and population interactions.(Hackl & Dubernet, 2019)

(Hackl & Dubernet, 2019) present a detailed study on compartment models which divides the population into compartments that represent the different stages of disease. SIR model which is the most standard approach of compartment model divides the population into three compartments: Susceptible (S), Infected (I), and Recovered (R). Susceptible contains individuals that has never been infected, most probably will catch the disease and this period is called infectious period. Infected individuals can spread the disease to the susceptible population. After a period of infection, individuals recover and move to the Recovered compartment, where they no longer contribute to disease transmission. An association of the nonlinear ordinary differential equations (ODEs) and transition rate to the mobility of agents between compartments represents the transmission of the virus in the SIR model. Although the model is easy and extendable because of the use of small parameter, (Hackl & Dubernet, 2019) argues that model restricts to population size being constant over time where demographics are not considered. The model's limitations are that it assumes a perfectly mixed population, ignores interpersonal interactions, and does not account for the spatial impacts of epidemic propagation.(Hackl & Dubernet, 2019)

In support of the compartment models, (Baggaley et al., 2005) discusses the use of mathematical models to predict the impact of antiretroviral therapy on HIV transmission and mortality. The study focuses on the use of two mathematical models -Dynamic and Linear models, which are the types

of compartment models. These models employ differential equations to characterize the rates at which people migrate between compartments. Dynamic model offers insights into ART use and its long-term effects by incorporating HIV transmission dynamics, in which the population's HIV prevalence affects the frequency of new infections. On the other hand, linear models are straightforward models which concentrated more on cost effectiveness analysis, ART treatment and individual health states. Since they frequently ignore the feedback loops that takes place in the transmission of infectious disease, they are appropriate for short term forecast. The model's results indicated that the dynamic models were efficient in capturing the complexities of long-term epidemiological impact by showing that ART can reduce viral load and infectiousness. Although the models offered better accuracy but there were difficulties in implementing and interpreting the models. (Baggaley et al., 2005)

In contrast to the traditional epidemic models, more accurate and realistic representation of population are considered in complex network model. One such network model is the metapopulation model which distinguishes type of population within each epidemic state allowing for a more nuanced analysis of HIV/AIDS disease spread dynamics across different demographic or geographic groups.(Hackl & Dubernet, 2019)The model defines individuals within the spatially structured subpopulation like cities, regions are characterized into class by their disease state: infected, susceptible, and recovered. These models account for local interactions and state changes due to infection dynamics. Additionally, the movement of individuals between subpopulations facilitates interactions and the spread of the disease across different locations. The model captures the complexity of human behavior especially in relation to mobility patterns and social network. However, it limits behavioral generalizations within the subpopulation(Hackl & Dubernet, 2019)

(Hadjandreou et al., 2007) discusses the study of an individual based simulator that models the HIV transmission and partnership concurrency called Mukwano consisting of 22 inputs and 18 outputs which is stochastic in nature. Mukwano simulates heterosexual partnership and HIV transmission where each individual has characteristics such as gender and date of birth, or change, such as HIV status, due to events like the start and end of sexual relationships that remain constant, these events occur stochastically, determined by probability distributions. To generate outcomes for a simulated population, individual characteristics are aggregated. The model accounts for births, deaths, partnership formation and dissolution, and HIV transmission using time-dependent rates. Furthermore (Hadjandreou et al., 2007) focuses on the use of emulator that utilizes the history matching, a calibration methodology for fast evaluation and producing large number of calibrated inputs that Mukwana fails to do so. The study's simulation results successfully highlighted

the significance of macrophage compartment in the development of HIV using the emulator for history matching. Virus ridden macrophage were discovered to grow gradually in number throughout the asymptomatic phase. However, history matching technique constraints in making probability statements about the simulator parameters hence it can only be viewed as a procedure that improves the efficiency rather than a direct competitor for other calibration models.

Agent Based Models in contrast overcome the limitations of compartment and metapopulation models by allowing interactions among individuals. Agent based models the epidemic spreading by tracking the disease progression through individuals and their contacts within social network like geographical areas like workplaces and schools. This method accounts for individual behavior variations and randomness, enabling the simulation of human actions through autonomous decision-making agents.(Hadjiandreou et al., 2007).The following provides a detailed description of agent-based modelling used in epidemic spread and HIV/AIDS transmission providing a base for the use of this technique in this project.

2.2 AGENT BASED MODELLING

Over the past few years, models for epidemiological forecasting of deadly illness like HIV/AIDS have been known to fail because individuals evolve their behavior to fit the epidemic. (Hunter et al., n.d.) believes that relying on equation based models to simulate the development of an infectious illness in a population is one of the primary reason of this kind of failure .The researcher also proposes that an alternate to the equation based model is agent based modelling .Within the last decade , agent based modelling paved its way into the modelling and estimation of epidemic spread.(Deeks et al., 2015; Hackl & Dubernet, 2019; Hadjiandreou et al., 2007).The computational study of social agents as dynamic network of independent interacting agents is known as agent based modelling.Agent based modelling is a tool to examine social systems from the perspective of complex adaptive systems.(Janssen, 2005)

Agent-based simulation models enable agents to interact with other agents and the environment based on a set of rules considering the model as an essential part of the epidemiology study. Each individual can be given different attributes and can make different decision when using an agent-based approach which captures the interaction and behaviour at individual level.(Hunter et al., n.d.) Agent based models approaches infectious diseases in two main ways: either by creating general models with adjustable disease parameters to simulate various diseases across different populations, or by focusing on specific diseases and outbreaks. General models, (Hunter et al., n.d.) are designed to be adaptable to multiple diseases with similar transmission methods, such as airborne

transmission. This flexibility allows researchers to use the same model to study different diseases and predict potential future outbreaks, aiding in policy decisions during pandemics.(Hunter et al., n.d.) describes that modelling for disease spread using agent-based modelling are divided into between host transmission and within host progression. A susceptible agent coming in contact with an infective agent occurred in host transmission. The within host progression component of a disease model simulates how, when an agent becomes infected, they move between the different states of the infection (for example, exposed, infective, and recovered). Both parts of the disease model are important in accurately simulating how a disease will spread. Furthermore (Hunter et al., n.d.) argues that transmission dynamic is a crucial factor in understanding how disease is spread among individuals. Diseases can be transmitted through human contact, contaminated food or water, or cross-species transmission, such as from mosquitoes to humans. When an agent encounters an infection source, a probability distribution determines whether transmission occurs. Factors influencing transmission include societal norms, transportation, environmental conditions, and individual behaviors. Agent-based models vary based on the disease's specific transmission dynamics. The following discusses models used in simulating the spread of HIV transmission.

(Gopalappa et al., 2016)designed and developed an agent based model Path 2.0 to incorporate dynamic HIV transmission based on sample of 10,000 people living with HIV (PLWH), who represented all men who have sexual intercourse with men (MSM) and heterosexuals living with HIV in the United States .In the model ,PLWH were characterized based on sex and age, sexual transmission risk group (heterosexual or men who have sex with men), disease stage (acute or non-acute), and their status within the HIV care continuum, from diagnosis to viral load suppression. Demographic and clinical characteristics were derived, and sexual behaviors associated with transmission were characterized based on data acquired. (Gopalappa et al., 2016) Path 2.0 model simulates the population of uninfected people stratified by gender and risk. The newly infected were transitioned into agent-based model when HIV transmissions occurred. When the Path 2.0 discovers that the newly infected individual has another infected partner, it avoids error by comparing the anticipated age and sexual orientation of the second infected partner to those of the previously identified infected individuals. To reflect the changes of the HIV infection and the evolution of the illness from diagnosis to viral load suppression, every HIV positive individual's disease specific values were updated in the model at each time step. The following stages of the illness: acute infection, asymptomatic infection, symptomatic infection/AIDS, and mortality were progressed through individual. The key findings of the model suggested that the risk of transmission was 9 times higher in the acute phase of HIV infection than in the non-acute phase. The highest

transmission rate came from individuals unaware of their HIV status in the non-acute phase. MSM was identified as the cause of most of the transmission which is greater than the rate among heterosexuals. Although the model was successful in capturing the complex dynamics HIV transmission, disease progression and impact of treatment due to its incorporation of real-world data, there were limitations in accuracy of some estimates because of the lack of data collected.

(Adams et al., 2018) study focuses the development of another agent-based model called Treatment of Infection and Transmission in Agent-Based Networks (TITAN) model, a dynamic model of HIV transmission to identify factors that influence the relationship between HIV acquisition in African American women and the widespread incarceration of African American men. The TITAN tool is considered as perfect for this analysis (Adams et al., 2018) as it has been effectively used to investigate how racial disparities in treatment occurs as preventive efforts among individuals who inject drugs, pre-exposure prophylaxis (PrEP) programs, and other intricate processes affect HIV transmission in a mature epidemic setting. The real-world trends of HIV transmission were calibrated as parameters in the TITAN model. The movement of men in and out of prison or jail were simulated to understand the interactions between incarceration and HIV acquisition in African American women. Discrete monthly steps and open population were used for running the model. The development of a population-based sexual HIV transmission network was illustrated when Sexual ties, or "links," between agents are established, maintained, or severed at each time step. The model carried out six scenarios with different important criteria that were thought to affect the impact of mass incarceration. The model results successfully calibrated the HIV prevalence and incidence trends for African American men and women residing in Philadelphia between 2005 and 2015. The model revealed that there is significant risk of HIV transmissions to African American women in the community due to the mass incarceration of African American men. However, the model was incapable of simulating the range of human experiences associated with incarceration such as poverty, stigma and drug relapse .(Adams et al., 2018)

The agent-based model called as the the Levers-of-HIV-Model is presented by (Vermeer et al., 2022) which simulates 15 years new HIV infection among MSM. The model serves as a tool to create different scenarios of change in the PrEP and ART cascades, called as levers to analyze trend in new HIV infections in Chicago. The implementation of evidence-based preventive and treatment interventions is the main focus of six levers, which are as follows: (1) linking individuals who test negative to PrEP; (2) linking HIV-positive individuals to ART; (3) adhering to PrEP; (4) using ART to suppress viral infection; (5) maintaining PrEP; and (6) maintaining ART. The simulations are

used as a support tool to inform policy makers by offering timely, cost-effective, and actionable information on how to implement strategies that fulfil GTZ goals. They can also be used to explore, tailor, and optimise implementation methods that target these levers of change. The HIV transmission and progression, health disparities, prevention and care, network dynamics and partner selection, and demographics are the five main modules that make up the LHM. Net Logo is the software that was used in this model for simulations. The scenario results from the study indicated that 75% reduction in HIV infections can be achieved by 2025 by combining high levels of PrEP linkage, retention, adherence and ART retention. The study's merit includes its emphasis on the cumulative effects of various treatments and its ability to provide customised insights for local public health efforts. Nevertheless, due to the focus on MSM in Chicago and certain assumptions made required to fill the data gaps, the findings are less generalisable and could affect the accuracy of the estimates. (Vermeer et al., 2022)

Agent based modelling can model much richer scenarios than previously established models. Agent based modelling are great tools to help understand how a population can be infected and how we should respond to an outbreak when new infectious diseases emerge. The fact that agent-based models provide for greater flexibility and design freedom in the simulation is one benefit they have over approaches like equation-based models. There is a price for this degree of freedom and flexibility: a lot of models in the literature suggest creating their own agent-interaction structure.

2.3 MACHINE LEARNING IN EPIDEMIOLOGY MODELLING

A branch of computer science and artificial intelligence that focuses on using data and algorithms to help AI mimic human learning processes and progressively become more accurate is called Machine learning. Without the explicit programming, computers can now learn with aid of machine learning. To make predictions machine learning uses complex algorithms and techniques. Machine learning algorithms are predominantly classified into 'Supervised' and 'Unsupervised'. In supervised learning, each observation yields the value of the result, often known as the dependent variable or its "label" in machine learning. Whereas in unsupervised learning, the algorithm attempts to identify naturally occurring correlations and groups within the data without considering any result or correct answer. Decision trees, support vector machine and random forest are some of the among the popular supervised learning algorithms. Unsupervised algorithms include clustering and dimensionality reduction. (Bi et al., 2019)

Machine Learning algorithms usage in medical research has steadily increased over the past decade. The algorithms are proved to assist in controlling the spread of infectious disease by aiding in the spatial and temporal prediction of the emergence and dissemination of infectious diseases. The capability of ML algorithms in analysing large, complex datasets, and identifying trends and patterns makes them well suited for predicting infectious diseases like HIV/AIDS involving multiple factors such as population demographics, environmental conditions and individual behaviours.(Santangelo et al., 2023).

Building on the strength of ML in medical research for infectious disease, (Bisaso et al., 2017) conducted a survey that utilized machine learning methodology in HIV/AIDS clinical research and medical care. The clinical research was classified as any web-based, hospital-based, social, clinical, imaging, genetic or molecular laboratory research involving the diagnosis and treatment of HIV/AIDS involving human participants or data from human participants. The purpose of the survey was to determine the trend in the usage of machine learning techniques, the particular ML and clinical tasks involved, the most popular ML methods, and their application paradigms and suitable data types. The findings of the survey revealed that there was an increase in the application of ML in HIV research from 1995 to 2016. The survey also observed that the most commonly applied ML methods was Support Vector Machined (SVM) followed by ANN and Logistic Regression. ANN and LR were the primary methods, but they were surpassed by SVM and Random Forest gradually in the observation of the survey. (Bisaso et al., 2017)

Implementation of machine learning algorithm in the HIV research was described in the study by (Bisaso et al., 2018) where ML was used to predict virological suppression in antiretroviral initiating HIV patients. The study proposed a methodology where three logistics regression approaches—Simple Logistic Regression (SLR), Multitask Temporal Logistic Regression (MTLR), and Patient-Specific Survival Prediction (PSSP) were used for prediction of the early virological outcomes in patients initiated with ART. The model was trained on the dataset to determine the feature coefficients and tested on the dataset to evaluate the performance of the model. The findings of the study indicated that MTLR model performed more effectively than others in key parameters such as calibration and discrimination. These findings suggest that MTLR was highly beneficial for this kind of prediction because of its ability to capture temporal dynamics. However, the model's complexity poses potential challenges in restricting its practical application in various healthcare environments.

Another study by (Balzer et al., 2020) depicts the application of machine learning algorithm to develop strategies that are required to prioritize individuals of higher risk of human

immunodeficiency virus acquisition for prevention services .HIV risk scores and their ability to seroconversions was constructed by gathering population level HIV data from rural Kenya and Uganda .Numerous HIV scores were created and verified in eastern and southern Africa to forecast HIV seroconversion within recognised risk categories, such as men who have sexual intercourse with men (MSM), African women , and Sero discordant couples.(Balzer et al., 2020) methodology follows by generating three risk scores to predict the 1 year risk of HIV acquisition using Super Learner method. It is ensemble of machine learning algorithms which includes logistic regression, stepwise logistic regression. The cross-validated risk scores were evaluated to assess different approaches aimed at intensified prevention. A person as categorized as "high-risk" if their risk score was higher than or equal to a cutoff determined by the score. The results of the machine learning in the study provided evidence for constructing population-level high HIV risk scores. The regression differentiated well between prevalent HIV cases and controlling. The study was subject to limitations with unreliable data and missingness in the data.

2.3 INTEGRATION OF AGENT BASED MODELLING AND MACHINE LEARNING

Adaptability of models and predictive accuracy can be enhanced by integrating machine learning with agent-based model. ML algorithms are being employed to improve the agent learning capabilities in agent-based modelling and create autonomous behavior. ML algorithms can calibrate ABM parameters using real-world data, and hence improving model fidelity. The decisions made by agents in agent-based model are guided by machine learning algorithms which can be used to create new behaviors throughout the simulation runs.ML algorithms are applied to ABMs in one of the following phases: (i) prior to running the ABM, (ii) during the run of the ABM, and (iii) after the ABM running to analyze the model outputs. In the first phase ML prepares input data by deriving parameter values based on empirical data. Whereas in the second phase, they train a learning algorithm by capturing ABM outputs during the simulation. ML calibrates and validates the ABM outputs in the last phase.(Abdulkareem et al., 2019)

(Angione et al., 2022) examined the use machine learning algorithm as a surrogate model for agent-based models to overcome the computational complexity and time demands in agent-based model simulations. The behavior of the original ABM to a high degree of accuracy is replicated using statistical model that were generated by using machine learning. Surrogate models are statistical models that effectively imitate the behavior of AMBs, reducing the number of resources needed for sensitivity analysis and policy modelling. Several ML algorithms such as artificial neural network

(ANNs), gradient-boosted trees (GBT), and Gaussian process emulators (GP) were compared to assess their performance. The findings of the study revealed that ANN performed better than other methods particularly in scenarios with large number of models runs offering high accuracy in accurately simulating the behavior of ABMs. While ANN model was efficient, its lack of interpretability was one of the primary obstacles to use it as a surrogate model. Also, it required significant computational resource for training.

Similar approach was used by (Larrie et al., 2021) to forecast sepsis disease progression by integrating ABM and ML. The study focused on reducing the computational complexity of ABMs by training ANN to act as surrogate model. Innate Immune Response agent-based model was employed to simulate the complex biological processes such as the body's inflammatory response to infection. Two types of artificial neural network (ANN) algorithms—Long Short-Term Memory (LSTM) and Multi-Layer Perception (MLP) were deployed to predict future cytokine levels and patient health status based on the agent-based model outputs. Overall health outcomes based on cytokine profiles were predicted using the regression network in the model. The ANN performed well in forecasting the patient health trajectories and approximating the time evolution of cytokine levels with MLP outperforming LSTM in predicting long term health outcomes. The model also reduced computational time compared to running full ABM simulations. While the model was a success, its ability to fully capture and predict disease progression was limited due the randomness in biological systems.

2.4 CRTICAL EVALUATION AND GAPS

According to the recent research, despite the usefulness of agent-based model in HIV/AIDs transmission, Agent based models are computationally expensive especially when dealing with population or environment. Limitation persists to their inability in fully capturing complex human behaviours. ABMs rely extensively on parameters to model the disease spread. However, selecting and fine tuning these parameters can be problematic since they depend on empirical data which may be incomplete or incorrect.

In this project, machine learning algorithm was integrated into the agent-based modelling to overcome the inherent limitations in simulating the effects of HIV /AIDS disease on human population and explore the management options. Machine learning can enhance the ABM by calibrating and validating the parameters based on historical data. Integrating ML into ABM provides the predictive analysis of long-term outcomes of HIV/AIDs transmission.

CHAPTER 3: METHODOLOGY

A multifaceted methodological approach is employed in this project to address state of the art in HIV/AIDS epidemic modeling and intervention strategies. The study simulates the HIV/AIDS epidemic effects on the human population and explores the management and mitigation options using agent-based modeling, concurrently machine learning techniques provide predictive analytics, enabling scenario analysis and optimization. This chapter outlines the data collection and preprocessing required for modeling, step-by-step design and implementation of the agent-based model, and the integration of the machine learning algorithm used to analyze the scenarios and predict outcomes of intervention strategies.

3.1 DATA COLLECTION AND PRE-PROCESSING

The data collection was focused on gathering comprehensive information on HIV/AIDS to analyse the spread and management using simulation and predictive modelling techniques. The data used in this project was obtained from Kaggle, a widely established platform for publishing datasets. (Kaggle)

Name	Description	Domain
Borough	Geographic region	Nominal
Gender	Gender of individuals in the dataset	Nominal
Age	Age of the individuals	Integer
Race	Ethnic background of the individuals	Nominal
HIV diagnoses	Number of HIV diagnoses recorded	Integer
Concurrent diagnoses	Diagnoses where HIV and AIDS were diagnosed simultaneously	Integer
Linked to care 3 months	Linked to HIV care within 3 months	Integer
AIDs diagnoses	Number of AIDS diagnoses recorded	Integer
Viral suppression	Percentage of viral suppression	Integer
Deaths	Number of deaths	Integer

The data set consists of 6005 observations and 12 variables. The key variables in the data set are depicted in the table above. Borough represents the administrative divisions within the geographic regions such as Manhattan, Brooklyn, Broncs, and Queens. The demographic variables which are Age, which indicates the age of the individuals, Gender is categorized into Male, Female, and Transgender and Race records the racial background of the individuals, are essential to understand HIV/AIDS transmission and outcomes. HIV diagnoses record the number of HIV diagnosed cases. Concurrent diagnoses represent cases where individuals are diagnosed with both HIV and AIDS. Similarly, AIDS diagnoses capture the records of individuals with AIDs. Linked to Care within 3 months variable tracks the percentage of people who start treatment immediately after being diagnosed and Viral Suppression indicates the proportion of individuals whose viral load has been lowered due to HIV treatment success. Lastly, the death variables indicate the deaths of individuals due to HIV/AIDS.

Data preprocessing and cleaning is a critical step in preparing the data set for simulation and analysis. During the data cleaning process, variables Age, Gender, and Race were converted into factor levels for enhancing analytical usability. Some of the variables that were not required were removed. No missing values were detected in the dataset. Additionally numerical variables with outlier values '9999' were considered as 'Na's as deleting them would alter the data. Duplicate values were identified and removed. Finally, after doing validity checks on the data, any remaining issues were examined, and a clean dataset was prepared for the agent-based model simulation.

3.2 AGENT BASED MODEL FOR HIV/AIDS

An agent-based model is a computational modelling technique for simulating the behaviour and interactions of autonomous agents, which can be individuals or collective entities like organizations or groups. This method is particularly suited for researching the transmission of infectious disease HIV/AIDs as it allows the simulation of intricate interactions between individuals and tracking of health status over time.

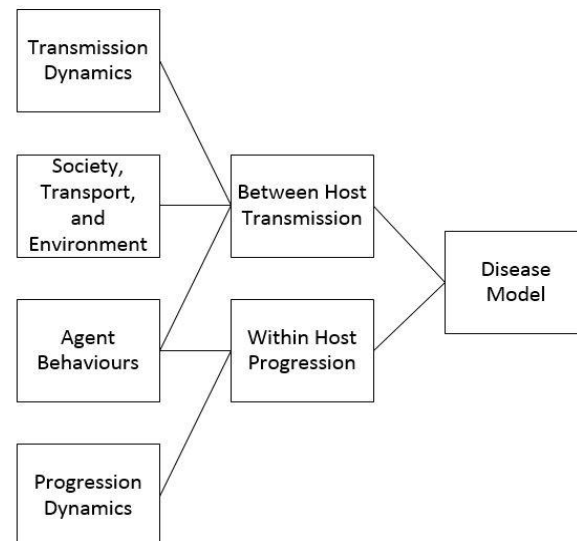


Figure 3.1 General structure of agent-based model for disease (Hunter et al., n.d.)

The above figure depicts the components of structure of the agent-based model which explains the factors such as transmission dynamics, society and environment, agent behaviour affects the between host transmission in a disease model. In this project an agent-based model was developed to simulate the transmission of HIV/AIDs disease within the population in Borough, the progression from HIV to AIDS, and ultimately deaths caused by HIV/AIDs. The model also simulates the effects of intervention strategies Viral suppression and Linked to Care on HIV/AIDs transmission. The ABM in this project follows a framework similar to the SIR (Susceptible, Infected, Recovered) model.

Software and Tools

The R software environment and R programming language were chosen for the development and execution of the agent-based model because of its strong statistical and analytical features. The extensive suite of packages and libraries and flexibility of R provides support for both agent-based modelling and data analysis. The R library packages dplyr, ggplot2 were used for data manipulation and summary statistics, reshape2 was used for reshaping data frames specifically for transforming facilitate plotting with ggplot2, and tidyr to facilitate analysis and visualization. The R functions sample(), runif(), which() were also used in the code.

Model Design:

Agents: The agents in this model represent individuals within the population of Borough and are subject to interactions that influence the transmission of HIV/AIDs during the simulation. Agents

are initialized by attributes id, Age which determines the susceptibility and behaviour, Gender that may affect transmission dynamics due to different risk profiles, Race, Viral Suppression and Linked to Care within months. The attributes excluding the id are variables derived from the HIV dataset. Each agent has a default attribute health status that determines how the agents interact with each other and their environment. The health status in this model includes states susceptible (not infected with HIV), infected (infected with HIV), AIDS (progressed to AIDS), and dead (has died due to AIDS). Initially, the agents are set to a 'susceptible' with positions (x,y) on the grid. Sample of the data was retrieved from the HIV dataset to initialize the agents.

Environment: The environment is a 2D grid of size 50x50 representing geographical locations where agents are located. The grid allows agents to move, interact with other agents within their immediate vicinity, and potentially spread the infection. The grid simulates spatial and social proximity affecting interaction patterns.

Transmission Rules: The core dynamics of HIV spread in the population is captured by the transmission rule. Proximity and random chance of transmission are critical factors for simulating the disease propagation through close contact between individuals. HIV is transmitted between agents during interactions if an infected agent comes into contact with a susceptible one. The probability of transmission is influenced by factors like interaction frequency and protective behaviours.

Behaviour Changes: The behavioural risk factors in this model represent the different ways agents interact and transmit HIV, such as through sexual activity, mother-to-child transmission, or sharing contaminated objects. Agents may alter their behaviour in response to interventions or changes in their health status, which in turn affects their risk of infection and progression to AIDS.

Simulation Process:

In this study simulations are run under three scenarios to model the spread of HIV/AIDS within a heterogeneous population. These scenarios are designed to investigate the dynamics of transmission and progression under varying conditions and demographic factors. A set of initial parameters, conditions and intervention techniques are applied to each scenario.

Simulation parameters: The key parameters that govern the simulation are:

- Infection probability determines the likelihood that a susceptible agent will contract HIV from an infected neighbour during each interaction. This parameter is set at 0.05 such that the interaction between an infected and susceptible has 5% chance of transmission

- Progression probability is the probability of progression of HIV to AIDS and is set at 10% per time step, following the natural course of untreated HIV infection.
- Death probability is set at 5% per time step determines the mortality rate due to AIDs related complication in the absence of an effective treatment.

Based on a combination of empirical evidence from pertinent studies , presumptions supported by previous research and factors specific to context of the modelled populations, these parameters were initialized and set.(Badri, 2006)The parameters infection probability was initialized based on the epidemiological studies of HIV (Boily, 2009) and progression probability was initialized based on the clinical studies that indicate an average of 5 -10% of untreated individuals progress to AIDS every year.(Morgan, 2002)

Simulation Loop: The simulation loop in this model run for 100 steps where each step represents the time unit. Hence 100 steps determine 100 days of the transmission of HIV/AIDS disease among the population. In view of the fact that HIV/AIDS is a disease characterized by relatively slow progression compared to other infectious diseases, a sufficient number of steps are required to observe the significant changes of the transmission.

Model Scenarios

Scenario 1: This is a baseline scenario that simulates the spread of HIV/AIDs within the population of 6000 and leading to the number of deaths caused by the disease without any interventions.

The R code in the appendix b depicts the simulation process of the scenario. Initially agents move randomly within the grid at each time step. To ensure that agents remain within the bounds of the grid, their positions are updated by adding a random value between -1 and 1 to x and y coordinates on the grid. The transmission of HIV occurs when an infected agent (if (agents\$health_status[i] == "infected")) interacts with neighbouring agents who share the same grid location. Susceptible agent will become infected based on the infection probability on random draw (runif) if it is located in the same position as an infected agent.

Agents who have become infected there is a probability of the HIV disease to progress to AIDS. The probabilities are derived from the dataset from variable (AIDs_diagnoses).If the random draw is less than the probability (if (runif(1) < aids_diagnosis_probability) ,the health status of the agent is changed into 'AIDS'.Agents with AIDS will lead to mortality if the random draw is less than death probability (deaths <- runif(length(aids_indices)) < death_probability) and the health status is changed to 'Dead'.

Scenario 2: The second scenario models the transmission of HIV/AIDs in the population over time with demographic variations. This simulation allows to examine the how HIV/AIDS impacts the different segment of the population. The agents are categorized by age, gender and race factors. The scenario is similar to first one with infected agents transmitting HIV to the neighbouring agents, progression to AIDs and ultimately to death of the agents if infected with AIDs. The simulation records the number of agents in different health status stratified by age, gender and race.

Scenario 3. The last scenario models the impact of intervention strategies Viral Suppression and Linked to care on the transmission of HIV, progression to Aids and death on population. Agents being virally suppressed and linked to care are initially set based on the dataset (hiv.df). During each simulation phase each agent moves randomly on the grid simulating possible interactions between the individuals. HIV can spread from non -suppressed (not having viral suppression load) infected agents to adjacent susceptible agents with a certain probability. Efforts to increase viral suppression rises linearly over time with agents linked to care having higher chance of viral suppression. Infected agents may progress to AIDs, but it is less likely if it is virally suppressed or linked to care. Deaths of agents with Aids is reduced if they are virally suppressed or linked to care.

The simulation of these scenarios runs for the number of time steps assigned. For each scenario at each step the health's status of the population tracks the number of susceptible, infected, AIDs, and dead agents are recorded. These results are stored in a data frame and then visualized using line plots and area plots to show the trends across the entire simulation duration.

3.3 Machine learning Algorithm for Predictive analysis of HIV/AIDS

The primary objective of employing machine learning algorithm in this study is to predict the future HIV/AID disease outcomes based on the historical simulation data generated by the HIV/AIDs agent-based model. The integration of machine learning algorithm with agent-based model can provide predictions and offer insights into future HIV/AIDS disease spread and efficacy of intervention by learning from past simulations. In this study machine learning algorithm is applied after running the agent-based model to analyze and calibrated the agent-based model outputs.

Random Forest Regression Algorithm was selected to predict the future HIV infection counts which is the dependent variable. It is a robust ensemble learning algorithm used for regression that operates by constructing a multitude of decision trees during training and outputs the mean prediction of each tree.

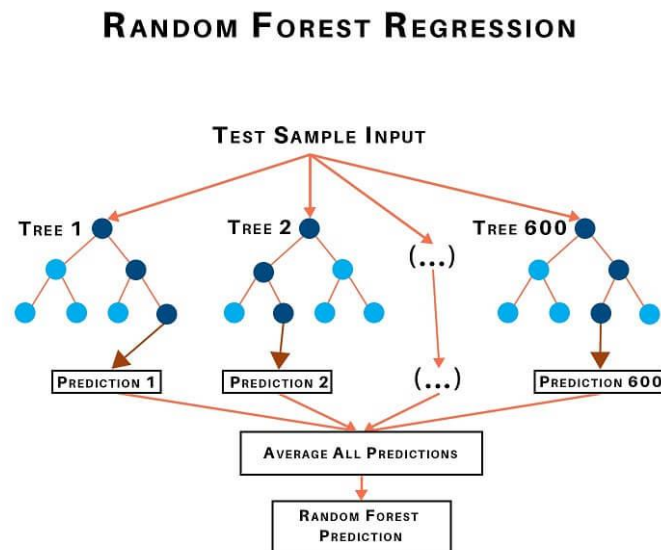


Figure 3.2 Random Forest Regression (Sharma, A. (2020))

As depicted in the figure 3.1, Random Forest takes prediction from each tree instead of just one decision tree and based on the majority of the predictions, it predicts the final output. More number of trees leads to more accuracy. It reduces overfitting by averaging the results of these trees. Due to its robustness, its ability to handle multiple input features such as the results from the simulation and manage complex, nonlinear relationships, Random Forest regression is a good fit for this task.

Data Preparation: The simulation results from the last scenario were derived to create features for the machine learning model. The simulation data results consist of the health status variables recorded at each time step which are infected, susceptible, aids, dead, viral suppressed, linked to care.

Feature Engineering: To create features that reflect temporal patterns and correlations important for predicting HIV infection counts, feature engineering was applied to the simulation results variables. In feature engineering lagging is the process of adding new features to a time series data sequence by adding specific number of periods (referred as “lags”) to the sequence. It involves transferring data for some variables backward in time by a predetermined number of time steps. When predicting future values based on past observations is the aim of time series analysis and modelling, this technique is especially helpful. Hence lagged versions of the key variables were

created to account for their values in previous time steps and to capture temporal dependencies and historical trends in disease progression.

- **susceptible_lag1:** Number of susceptible individuals at the prior time step (t-1).
- **infected_lag1:** Number of infected individuals at the prior time step (t-1).
- **aids_lag1:** Number of individuals with AIDS at the prior time step. (t-1)
- **dead_lag1:** Number of deceased individuals at the prior time step. (t-1)

Data Splitting: The simulation results data is divided into training and testing sets with 70% of the data used to train the model and 30% of the data used to evaluate the model's performance.

Model Training Process: A formula is defined for predicting the dependent variable infected individuals. 'infected $\sim$ susceptible_lag1 + aids_lag1 + dead_lag1 + viral suppressed + linked_to_care + step'. The Random Forest model is trained on the training data using the random forest function 'random Forest', with the following parameters formula, number of trees set to 500 and variable importance assessed to understand the influence of each feature. The random forest model predictions of number of infected individuals are done test data.

Model Evaluation: The model's performance is assessed using the metrics Mean Absolute Error (MAE) and Mean Squared Error (MSE). The number of errors in a set of predictions is measured by MAE. It provides the average absolute difference between the predicted and actual values of the number of infected individuals in this model. MSE measures the average of the square of errors in the model.

Model Visualizations: Variable Importance plot is used to plot the importance of each feature in the model that shows the variable having most influence on the predictions. Model's performance is visualized by plotting the actual and predicted infections count over time to compare the predicted values against the actual values using the functions `geom_point()` and `geom_line()`.

The methodology of this project focused on simulating the transmission of HIV/AIDS using the Agent based model under three different scenarios ,first one is a baseline scenario of general transmission , second scenario depicted the spread of the disease across stratified demographic factors age, gender and race and the last scenario explored the impact of intervention strategies on the disease trajectory .Additionally , Random Forest Algorithm was integrated and applied to the agent based model outputs to determine the future infection counts.

CHAPTER 4: RESULTS AND DISCUSSION

This chapter presents the results and conclusions derived from two primary components of the study conducted in the methodology: the agent-based model of HIV/AIDs disease transmission and the implementation of machine learning algorithm for predictive analysis. The goal of this research was to simulate the HIV/AIDS spread within a population in Borough and to explore the implications of various management strategies on the disease progression. Furthermore, machine learning techniques were deployed to forecast future disease outcomes and trends based on the simulation data. The results obtained from these methods were illustrated using tables, visualizations and performance metrics. Finally, this section concludes with discussion of findings, relating with the existing literature findings, and discussing limitations of this study and future work

4.1 AGENT BASED MODEL RESULTS

The agent-based model simulated the interactions of 6000 agents over 100-time steps, representing the spread of HIV/AIDs and the course of the disease. The simulation integrated numerous characteristics of the agents, including age, gender, race and health state to reflect realistic situation of the disease transmission and progression. The findings shed light on how HIV/AIDS changes over time in a population that is mimicked as well as the outcomes of various therapeutic techniques. The findings of agent-based simulations are divided into scenario wise

Scenario 1 Results: This scenario explored the general transmission of the HIV/AIDS disease in the population, progression to deaths and ultimately deaths.

Step	Susceptible	Infected	AIDS	Deaths
1	1384	1462	2997	157
2	1334	1457	2861	320
3	1332	1484	2735	459
4	1280	1497	2627	596
5	1248	1501	2507	744
6	1216	1509	2419	856
7	1175	1519	2322	984
8	1141	1531	2242	1086
9	1108	1539	2156	1197
10	1062	1554	2068	1316
.....				
50	295	1773	452	3480
51	289	1774	438	3499
52...	281	1775	425	3519
99	53	1844	95	4008
100	52	1844	91	4013

Figure 4.1 Table showing health status results for scenario 1

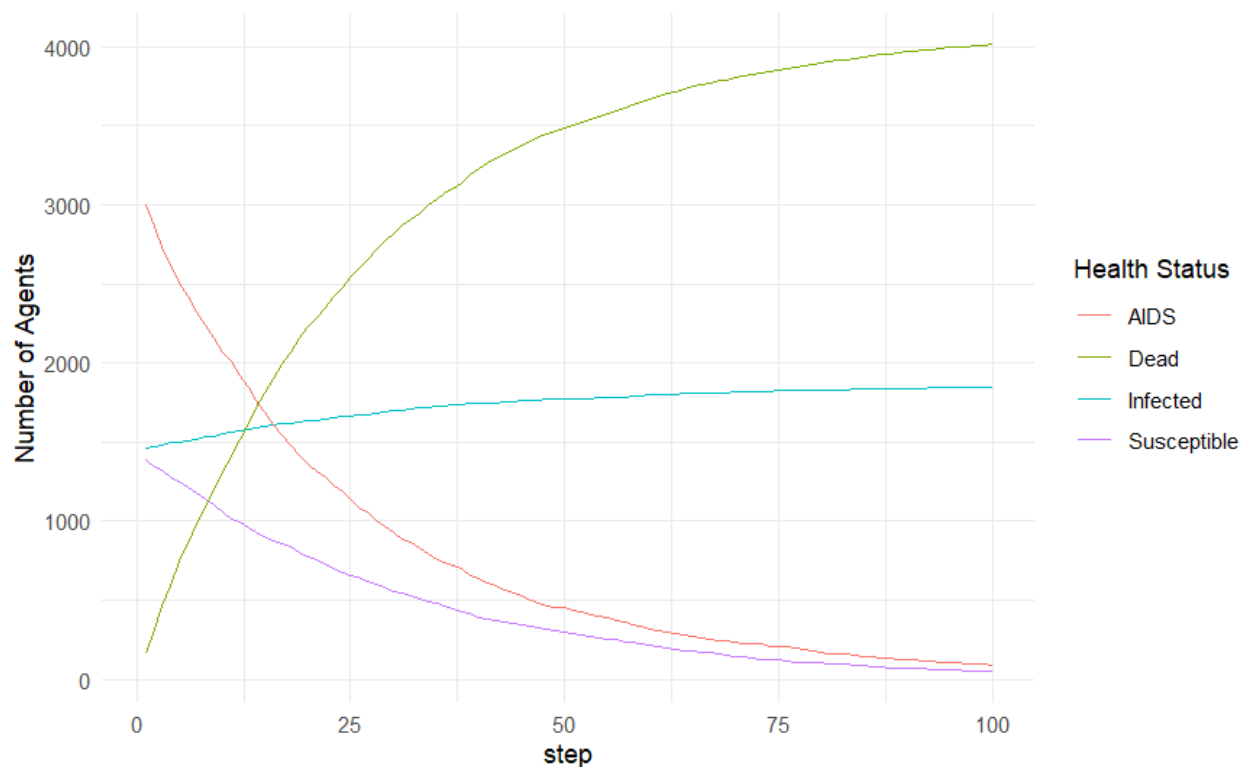


Figure 4.2 Line plot showing health status for scenario 1

The figure 4.1 table depicts the outputs of the simulation of the scenario. The columns represent the state of the population at a specific time step in the simulation.

Step: Current time step in the simulation

Susceptible: Number of agents who are still susceptible to HIV and not yet infected

AID: Number of agents progressed to AIDS

Dead: Number of agents who are dead.

The line plot in figure 4.2 is a time series line plot that tracks the number of agents in different health states over 100-time steps where x-axis represents the simulation steps and y -axis represents the number of agents. From the analysis of the plot, most agents about 1384 are in the susceptible state at the beginning of the simulation. As the simulation progresses, the number of infected agents increases while the susceptible agents decrease illustrating the spread of HIV within the population. Few agents emerge to progress to AIDS and subsequently dies. This is due to initialization of the variable AIDs diagnose from the dataset indicating few of the individuals of the population were already diagnosed with AIDS. At around step 50, the number of infected agents reaches a peak. As more individuals begin to progress, the number of agents with AIDS is also gradually rising at this point. As the simulation approaches its last steps, most of the agents has either died or become infected. The number of susceptible and AIDS has noticeably decreased suggesting that agents who contracted AIDS has died. As more infected people become sick with AIDS or pass away, the number of infected people stabilizes and even shows a slight decrease. This pattern illustrates how the epidemic naturally affects the population over time with the transmission of HIV eventually reducing the susceptible population and stabilising the overall number of infected people.

Scenario 2 Results: The HIV spread in the population segmented by demographic variations was explored in this scenario.

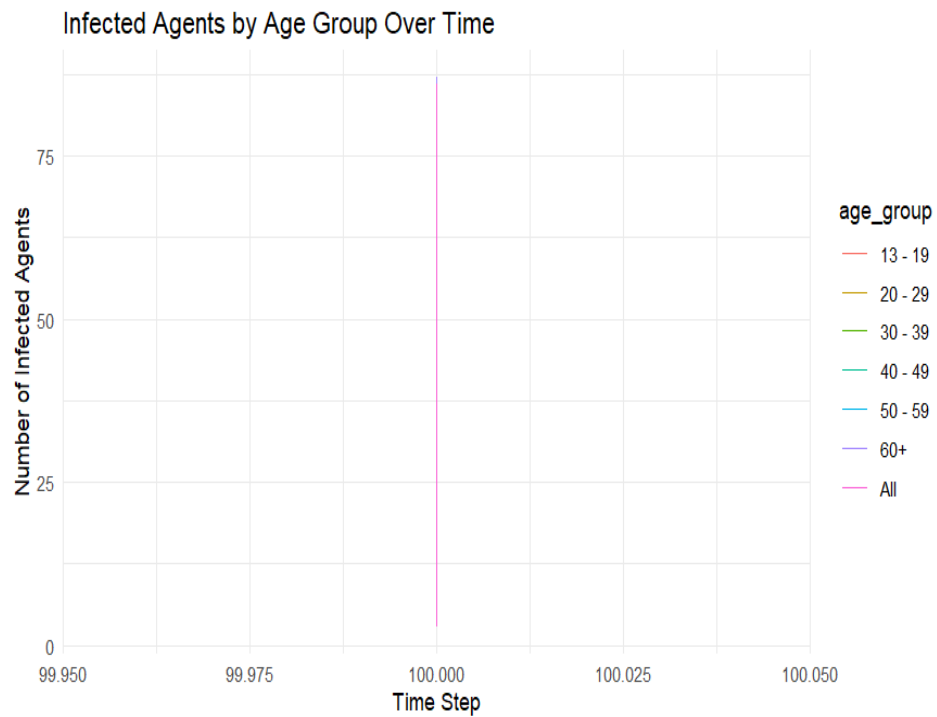


Figure 4.3 Line plot showing infected agents by age group

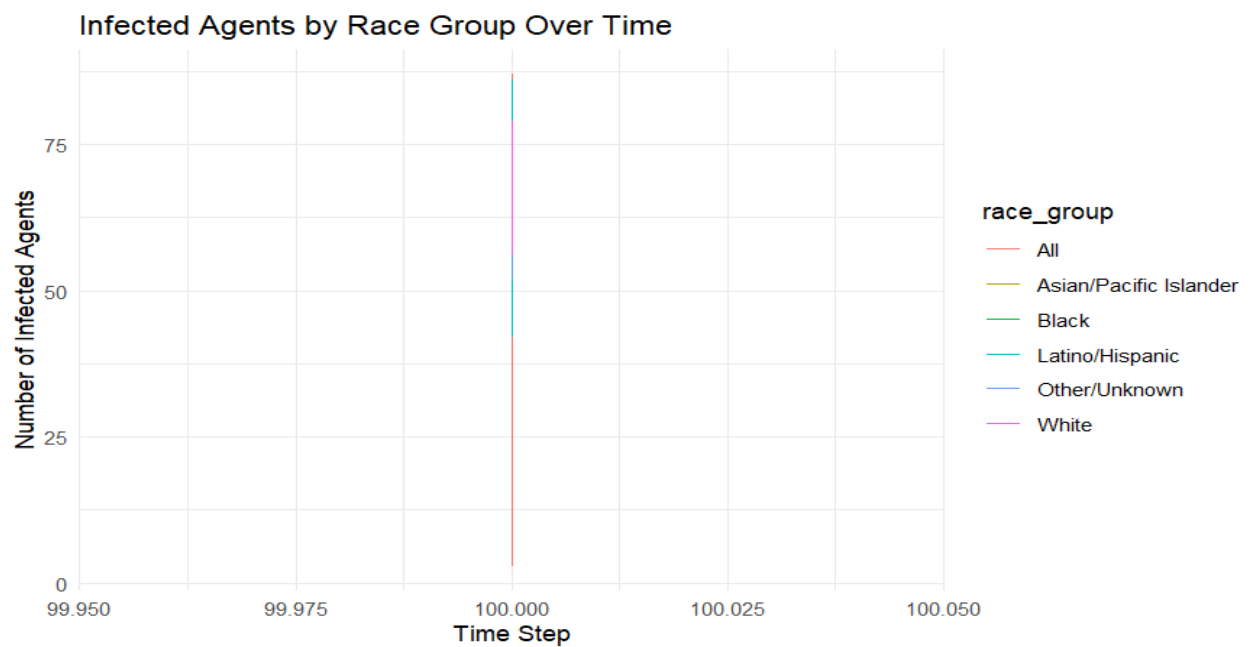


Figure 4.4 Line plot showing infected agents by Race group

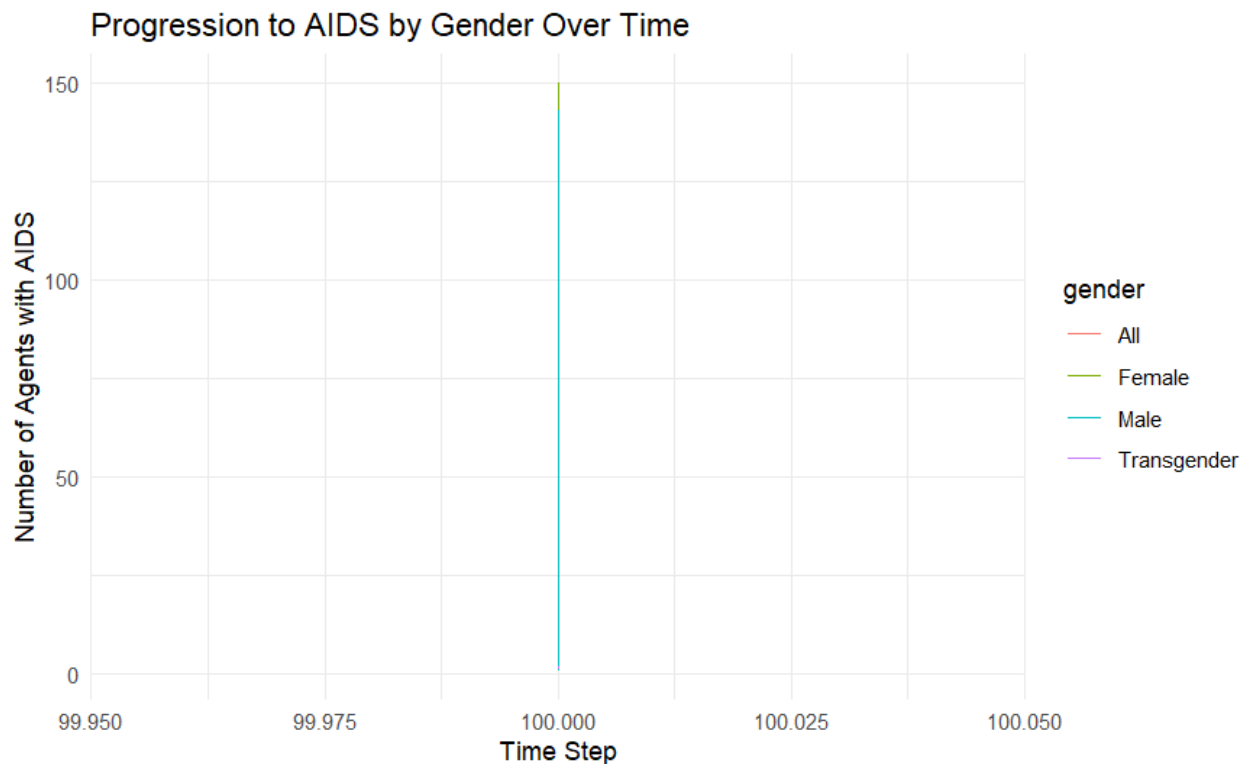


Figure 4.5 Line plot showing progression to AIDs by gender.

Figure 4.3 and Figure 4.4 presents the line plots depicting the number of infected across different age and race group over simulation time steps. Most of the data points have concentrated on around step 100 causing the straight line in the plot. The plot indicates that HIV infection affects all age group, with no significant differences in infection rates by age at time step 100. The line in the plot showing infected agents by race group are overlapping indicates that all race groups have similar number of infected agents. Hence it suggests that there was no significant difference in infection rates among different race groups in the population of Borough.

Figure 4.5 illustrates the progression from HIV to AIDS, broken down by gender. The plot reveals that a higher proportion of agents who progress from HIV to AIDS are male compared to the fewer proportion of females.

Scenario 3 results: The last scenario explored the impact of intervention strategies Viral Suppression and Linked to Care on the transmission of the HIV/AIDS disease spread.

Step	Susceptible	Infected	AIDS	Deaths
1	1517	3848	607	28
2	1513	3295	1110	82
3	1511	2855	1479	155
4	1506	2490	1768	236
5	1502	2157	2006	335
6	1498	1859	2218	425
7	1497	1629	2355	519
8	1494	1473	2392	641
9	1493	1315	2444	748
10	1491	1181	2466	862
.....				
50	1478	79	834	3609
51	1478	73	811	3638
52...	1478	68	786	3668
99	1478	6	183	4333
100	1478	5	178	4339

Figure 4.6 Table showing health status results for scenario 3

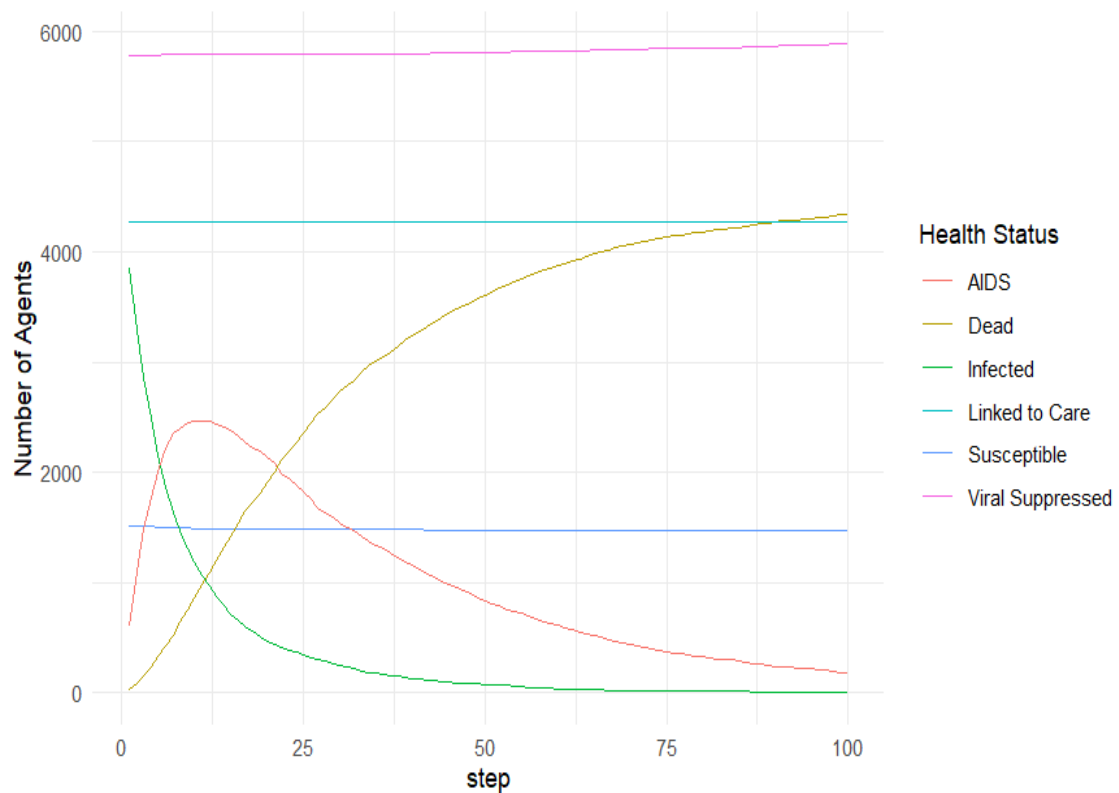


Figure 4.7 Line plot showing health status for scenario 3

The results of the simulation of the transmission of HIV/AIDs under the intervention strategies Viral Suppression and Linked to care are represented by the above line plot. It can be observed that there is an initial increase in the number of infected agents at the beginning of the simulation but decreases rapidly over time. This decline reflects the effects of viral suppression and the progression of some individuals to AIDs. The constant line for the susceptible population indicates that the number of individuals at risk for HIV infection remains unchanged throughout the simulation. This suggests a stable population due to impact of Viral suppression or access to linked care within 3 months where new susceptible are not introduced.

The number of individuals progressing to AIDs line starts at zero but rises initially, peaking around 25th time step before gradually decreasing. This trend demonstrates the positive impact of the interventions in slowing the progression. The steady increase in the dead state line reflects the ongoing impact of the disease, with the line eventually plateauing, indicating a reduction in mortality rates over time. This smoother curve, compared to an earlier scenario 1, suggests that while viral suppression may not have been strong enough initially—resulting in early deaths—it became more effective as the simulation progressed.

Viral suppressed line remains constant throughout the simulation indicating that a steady number of individuals maintains viral suppression possibly due to effective treatment. Similarly Linked to care line remains fairly stable over the simulation suggests that individuals linked to care has had positive effects from treatment.

From the results of the simulation of scenario 3 , it can be analyzed that Viral suppression and linkage to care have had a significant positive impact on controlling the progression of HIV and reducing mortality .The stabilization of AIDs cases and deaths over time reflects the effectiveness of the these strategies .However the initially high infection and death rates suggest that these interventions had more profound impact in later stages of the simulation

4.2 RANDOM FOREST MODEL RESULTS

Random forest regression algorithm approach was utilized to predict the future disease outcomes such as the infection rates of HIV/AIDS. The model was trained and tested on the results retrieved from the simulation of the agent-based model which includes the health status of the disease. Key metrics including the mean absolute error (MAE), mean squared error (MSE) was used to assess the model's performance. Additionally, percentage of variance explained was used to evaluate the

accuracy and reliability of the infection rate prediction

```
Call:
  randomForest(formula = results_formula, data = trainData, importance = TRUE, ntree = 500)
  Type of random forest: regression
    Number of trees: 500
No. of variables tried at each split: 2

  Mean of squared residuals: 22185.6
    % Var explained: 94.83
Mean Absolute Error (MAE): 14.73527
```

From the above random forest model results, the mean absolute error is the absolute difference between the predicted infection rates and the actual infection rates, a value of 14.7 suggests that model's predictions are off by 15 infections. This implies that the predictions are fairly close to the actual infections rate. The percentage of variance indicates that model explains 94.83% of variance in infection rates. The high percentage suggests that model is a good fit on the trained dataset and captures almost all the variability in the data. Mean squared error is the squared differences between the predicted and actual infection rates. The value of 22185 which is squared value of the original values implies that model's predictions are aligned with the actual outcomes.

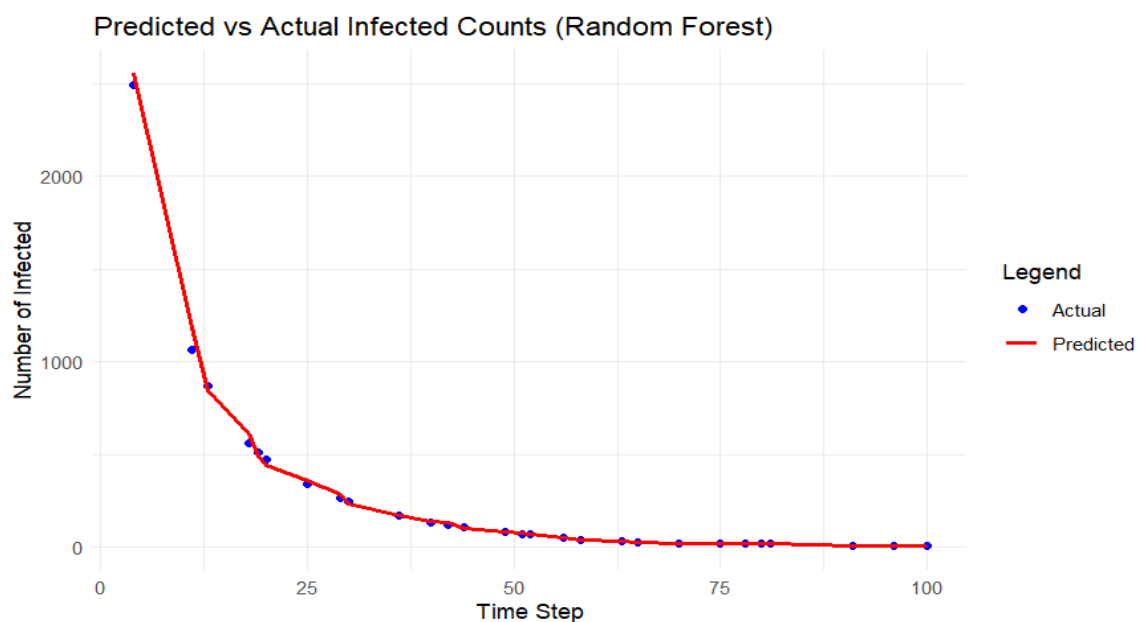


Figure 4.8 Line plot showing predicted vs actual infected counts

The above figure illustrates the performance the Random Forest model in predicting the number of HIV infected individuals over a series of time steps. It shows the comparison of the actual infected individual represented by blue dot and predicts infected individual represented by red line. The red line aligning with the blue dots indicates that throughout the series the model is accurately predicting the number of infections over time. There is a decline in the infection rates which is due to the impact of intervention strategies which was applied in scenario 3.

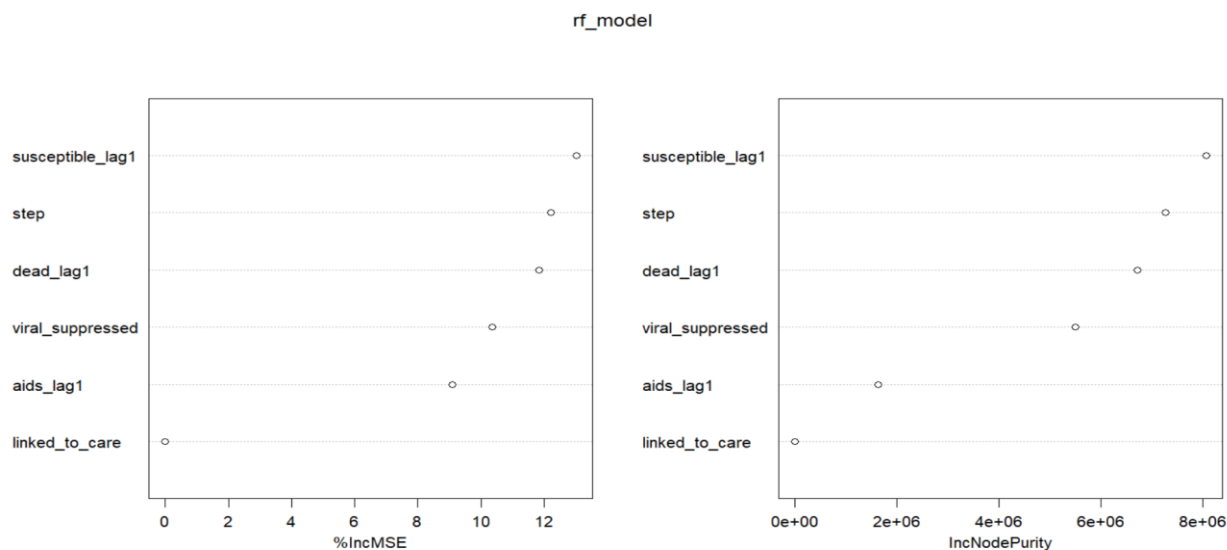


Figure 4.9 Variable importance plot of Random Forest

The variable important plot in the above figure 4.6 help in understanding the most important features in predicting infections. It is evident from the plot that susceptible is the most important feature while linked to care being the least. The next important variable is ‘Dead’ followed by the Viral suppression suggesting that these variables are key factors in analyzing future infections and understanding the transmission of HIV.

Overall, given the model’s high predictive accuracy, it can be inferred that the model provides a robust framework to forecast future HIV infections rates and the progression to AIDS in the population. Future infection rates are mostly influenced by factors susceptible, dead and viral suppression. The importance of viral suppression and linkage to care indicates that there is a significant impact of intervention strategies on the transmission of HIV/AID disease. These insights

are useful in understanding how the different factors influence the spread or control of HIV/AIDS in the simulated environment.

4.3 DISCUSSION OF FINDINGS

This study's findings shed significant insights into the dynamics of HIV/AIDS transmission and the effects of different intervention strategies to explore the best management option in controlling the disease. The agent-based model's scenario wise simulations disclose several critical patterns in the spread and progression of HIV to AIDs. The model exhibited an epidemiological curve, with HIV infections rising initially, a rapid progression from HIV to AIDS causing a peak in the number of AIDS cases and then a subsequent decline as mortality rates increased. This pattern aligns with the findings (Gopalappa et al., 2016) who identified similar trends demonstrating that the risk of transmission of HIV infections in acute phase was 50 times as high as compared those in non-acute phase. Similarly, the simulation also observed that the disease's initial exponential growth is followed by a plateau as the number of susceptible diminishes when the disease progresses to next stages. The segmentation of demographic factors in the simulation which analyzed infection rates by age and race highlighted that HIV/AIDS impacts all age groups relatively uniformly. Contrary to the uniformity observed in age and race factors, the simulation revealed a notable gender disparity with higher proportion of males progressing the disease from HIV to AIDS. This observation is consistent with findings by (Adams et al., 2018) which states that HIV/AIDS prevalence in men was higher with values 2397 per 100,000 compared to female with values 1857 per 100,000 and with average annual HIV incidence was 75 per 100,000 for men and 64 per 100,000 for women. This also aligns with another study's findings by (Andrianakis et al., 2017) which describes that high proportion of men belonged to the groups with elevated sexual activity group and contract rate which contributed to their increased risk of developing AIDS.

Introduction of intervention strategies Viral suppression and linkage to care in the simulation demonstrated an impact of reduction in both HIV transmission and deaths over time. This results are supported by studies as such as those by (Vermeer et al., 2022) which states that initiation of PReP and ART treatment increased the Viral suppression subsequently leading to decrease of disease progression and improve survival rates .However , while the results of the simulation reflects the benefits of the viral suppression and linked to care , the model's initial high infection rates and and mortality rates suggest that these interventions were more effective in later stages of the simulation. Hence this confirms that the best management or mitigation option for HIV/AIDS is the

early ART treatment initiation that can result in better health outcomes and reduced transmission rates.

The implementation of Random Forest model to predict future disease outcomes such as the infection rate in this model demonstrated high accuracy. The results of the model are consistent with similar studies by (Bisaso et al., 2017) which reveals that Random Forest models have been effective in predictive analytics for epidemiological data. Techniques like Random Forest are particularly effective in tackling nonlinear correlations and interactions in health data. The model's high percentage of variance explained indicates that it effectively captures the intricate dynamics of HIV transmission and the effects of interventions. The variable importance analysis revealed that number of susceptible individuals and viral suppression were most predominant factors in predicting infection rates. The high degree of significance of "susceptible" highlights the crucial role of the pool of at-risk individuals in determining future infection rates which emphasizes the importance of preventive measures to reduce the susceptible population. The incorporation of feature engineering method utilizing the lagging features to improve the prediction accuracy enhanced the model's capacity by providing a more comprehensive understanding of future disease trajectory. The alignment between the actual and predicted infection rates as demonstrated in the variable importance implies that the model can accurately anticipate future infections surges or declines in the HIV transmission based on patterns and interventions. The prediction of future infection rates by the model underlines its potential use in HIV infection trends and evaluating the effects of different intervention approaches. Infection predictions indicates that future infections will occur but can be significantly mitigated by effective interventions such as ART initiation which increases the viral suppression and increased access to care. Accurate prediction helps in public health planning and resource allocation which leads to more efficient response tactics. Health authorities can more effectively prepare for potential surges of the HIV/AIDS disease by anticipating future infections. The effectiveness of interventions Viral suppression and linkage to care helps public health programs to emphasis on increasing universal access to ART treatment and care, encouraging early diagnosis and testing of HIV and guaranteeing adherence to treatment plans.

Limitations And Future Work

The study to simulate the effects of HIV/AIDS on human population using agent-based model and explore the management options using integrated machine learning to predict future disease outcomes is subject to limitations that should be addressed. The agent-based model in this project assumes a basic set of behavior, interactions patterns between the agents and transmission probabilities. Complex behavior and socioeconomic conditions that may affect the rates and

development of transmission are not taken into consideration. Since the data is retrieved from a public platform, it may have limitations of sampling bias, incomplete records, or outdated information. The model also assumes that all the individuals in the population has access to linkage care which might contradict the real-world scenario. The Random Forest model data relies on the past simulation data which might not fully capture the real-world variations and complexities.

Future work could be incorporating more factors such as the detailed individual interaction patterns, socioeconomic status, educational level, real world statistics of access to linkage care in the agent-based model. Incorporating reliable real time epidemiological data and behavioral factors to refine predictions and enhance the model's generalizability could benefit future studies. Employing the method integrating machine learning with agent based for large scale database and organization is another future implementation.

CHAPTER 5: CONCLUSION

This project presented the simulations of effects of HIV/AIDS disease on human populations using agent-based modelling and explored the management options of the interventions required to curb the disease and forecasting disease outcomes using machine learning algorithm. An existing literature on the epidemiology models, agent-based models and machine learning algorithm used in the HIV/AIDS transmission was immensely reviewed and analyzed. The agent-based model was designed and developed to simulate the dynamic HIV/AIDS transmission among the individuals of the population by initializing the agents in the model with demographic and health status attributes. The agent-based model addressed the complexities of HIV transmission by incorporating data such as the intervention strategies Viral Suppression and linkage care. The scenarios of the simulation demonstrated the general HIV spread, progression of the disease to AIDS, deaths caused by the disease and impact of the intervention strategies on the disease transmission. The scenario results were depicted using line plots which showed variations of the disease transmission. The health status results were recorded and stored in data frame for further requirements. This approach provided a distinct flexible framework for comprehending the ways in which individual behavior and ultimately disease outcomes might be impacted by population level intervention. The key strength of this project was the integration of machine learning algorithm for enhancing the predictive accuracy of the agent-based model. Random Forest Regression algorithm was applied to output of the simulation data generated by the agent-based model to forecast the future infections count. The model incorporated lagging a feature engineering method which captured the temporal dependencies of the disease progression also leveraging the key variables such as the number of infected, susceptible, and deceased. The model's performance confirmed the robustness of the machine learning algorithm in handling multiple input features and nonlinear relationships. The application of Random Forest algorithm was successful in calibrating and validating the model's ability to forecast future transmission trends. The discussion of findings explored importance of the both the methods in understanding the dynamics of HIV/AIDs disease and further related with the existing findings. It also examined the management options such as the intervention strategies viral suppression and linkage care to prevent the transmission. The importance of these strategies helps public health to necessitate decision making process by reserving resources and requirements in reducing the HIV/AIDS disease. Overall, this study emphasized the importance of combining simulation-based strategies such as ABM with data-driven techniques machine learning. With the help of this hybrid method, HIV/AIDS transmission dynamic can be analyzed more precisely giving

Simulating the effects of HIV/AIDs on human population and exploring the management option using Agent Based Modelling and Integrating Machine learning for Predictive Disease Outcome Analysis

decision makers useful information about which intervention efforts are most likely to be successful. Although the project demonstrated the potential of this hybrid approach, certain challenges were faced during the implementation of this project such as the data availability constraints on the epidemiology and demographic factors and the computational intensity of ABM.

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APPENDIX A: Ethical Approval



College of Engineering, Design and Physical Sciences Research Ethics Committee
Brunel University London
Kingston Lane
Uxbridge
UB8 3PH
United Kingdom
www.brunel.ac.uk

12 July 2024

LETTER OF CONFIRMATION

Applicant: Ms Keerthana Kakkiriken Poothatta

Project Title: Simulating the spread of swine flu disease on human population using agent based modelling and exploring management and mitigation strategies using machine learning algorithms

Reference: 49008-NER-Jun/2024- 51586-1

Dear Ms Keerthana Kakkiriken Poothatta

The Research Ethics Committee has considered the above application recently submitted by you.

This letter is to confirm that, according to the information provided in your BREO application, your project does not require full ethical review. You may proceed with your research as set out in your submitted BREO application, using secondary data sources only. You may not use any data sources for which you have not sought approval.

Please note that:

- **You are not permitted to conduct research involving human participants, their tissue and/or their data. If you wish to conduct such research (including surveys, questionnaires, interviews etc.), you must contact the Research Ethics Committee to seek approval prior to engaging with any participants or working with data for which you do not have approval.**
- The Research Ethics Committee reserves the right to sample and review documentation relevant to the study.

If during the course of the study, you would like to carry out research activities that concern a human participant, their tissue and/or their data, you must submit a new BREO application and await approval before proceeding. Research activity includes the recruitment of participants, undertaking consent procedures and collection of data. Breach of this requirement constitutes research misconduct and is a disciplinary offence.

Good luck with your research!

Kind regards,

A handwritten signature in black ink, appearing to read 'Simon Taylor'.

Professor Simon Taylor

Chair of the College of Engineering, Design and Physical Sciences Research Ethics
Committee Brunel University London

APPENDIX B

```
---
title: "Dissertation Project:Simulating the effects of HIV/AIDS disease on
human population using agent based model and exploring the management
option ,and integrated machine learning for predictive analysis of future
disease outcome"
output: html_notebook
---
```

```
#Loading the necessary libiraries required for the modelling
```

```
```{r}
library(reshape2)
library(dplyr)
library(validate)
library(ggplot2)
library(scales)
library(papaja)
library(GGally)
library(caret)
library(doParallel)
library(randomForest)
library(knitr)
```

```
```
```

```
#Loading the original dataset as data frame
```

```
```{r}
hiv_df <- read.csv("DOHMH_HIV_AIDS_Annual_Report.csv")
```
```

```
#Getting the dimensions of the data
```

```
```{r}
nrow(hiv_df)
ncol(hiv_df)
```
```

```
#Exploring the dataframe
```

```
```{r}
View(hiv_df)
head(hiv_df)
```
```

```
#Exploring the variables
```

```
```{r}
str(hiv_df)
summary(hiv_df)
```
```

Simulating the effects of HIV/AIDs on human population and exploring the management option using Agent Based Modelling and Integrating Machine learning for Predictive Disease Outcome Analysis

```
#.....Data Processing and Cleaning....
#.....#

#Removing variables that are not required for further modelling and
analysis
```{r}
hiv_df[,
c('Year','UHF','Death.rate','HIV.diagnosis.rate',"AIDS.diagnosis.rate",'Non
.HIV.related.death.rate','PLWDHI.prevalence')] <- list(NULL)
```

#Renaming columns names
```{r}
hiv_df <- hiv_df %>%
 rename(

 HIV_diagnoses = HIV.diagnoses,
 Concurrent_diagnoses = Concurrent.diagnoses,
 Linked_to_care_3_months = X..linked.to.care.within.3.months,
 AIDS_diagnoses = AIDS.diagnoses,
 Viral_suppression = X..viral.suppression,
 Deaths = Deaths,
 HIV_related_death_rate = HIV.related.death.rate

)

```

#checking the renamed columns
```{r}
colnames(hiv_df)
```

```{r}
str(hiv_df)
```

#converting categorical variables with defined levels into factors for
accurate representation and enhancing analytical usability and explore for
any inconsistencies in the levels
```{r}
hiv_df <- hiv_df %>%
 mutate(

 Gender = as.factor(Gender), # Categorical
 Age = as.factor(Age), # Categorical (if
age ranges)
 Race = as.factor(Race), # Categorical

)
```

```{r}
table(hiv_df$Age)
table(hiv_df$Gender)
table(hiv_df$Race)
```

#Generating frequency table for categorical variables
```

Simulating the effects of HIV/AIDs on human population and exploring the management option using Agent Based Modelling and Integrating Machine learning for Predictive Disease Outcome Analysis

```
```{r}
cat_col_names <- c('Gender' , 'Age', 'Race')
hiv_cat_table <- apply(hiv_df[, cat_col_names], 2, table)
hiv_cat_table
```

#Note gender, age , and race has 'All' category and Race has unknown
category

#Proportional table for gender,age and race
```{r}
cols<- c('Gender','Age','Race')
cat_prop_table <- lapply(hiv_df[,cols], function(x) prop.table(table(x)))
cat_prop_table
```

#Applying validation rules to assess the variable constraints
```{r}
val_rules<- validator(non_neg_Year = Year>0 & Year<2025,
 ok_Age = is.element(Age,c('13 - 19', '20 - 29', '30
- 39', '40 - 49', '50 - 59', '60+', 'All')),
 ok_Gender=
is.element(Gender,c('Female','Male','Transgender','All')),
 ok_Race= is.element(Race,c("Asian/Pacific
Islander","Black","Latino/Hispanic","White","Other/Unknown",'All')),
 ok_HIV_diagnoses = HIV_diagnoses != 99999,
 ok_Concurrent_diagnoses= Concurrent_diagnoses !=
99999,
 ok_Linked_to_care_within_3_months=
Linked_to_care_within_3_months != 99999,
 ok_AIDS_diagnoses = AIDS_diagnoses != 99999,
 ok_Deaths = Deaths != 99999

)

qual_check <- confront(hiv_df,val_rules)
summary(qual_check)
```

#Missing value analysis
```{r}
count_miss<- colSums(is.na(hiv_df))
count_miss
```

#There are no missing values identified in the dataset

#Replacing 99999 with Na values in numerical columns
```{r}
hiv_df <- hiv_df %>%
 mutate(across(where(is.numeric), ~ replace(., . == 99999, NA)))
```
```



```
```{r}
table(hiv_df$Gender)
table (hiv_df$Age)
table(hiv_df$Race)
```

#Exploring any duplicate values
```{r}
duplicated_df <- hiv_df[duplicated(hiv_df),]
paste("Number of duplicated records found : ", nrow(duplicated_df))
```

#Removing the duplicated records
```{r}
hiv_df_dedup<- unique(hiv_df)
dim(hiv_df)
```

#Checking the cleaned dataset
```{r}
str(hiv_df)
View(hiv_df)
```

#.....Agent Based
Modelling.....

```{r}
Defining agent based model parameters

population_size <- 6000 # Population size representing agents, the
population is initialized to number of record in the data set
grid_size <- 50 # Size of the 2D grid

Setting the seed for productability
set.seed(42)

Sampling data from the hiv dataset to assign the initial health status
required for the modelling

sample_indices <- sample(1:nrow(hiv_df), population_size, replace = TRUE)
samplehiv_data <- hiv_df[sample_indices,]

Initializing agent's attributes using data frame based on the sampled
data from the original data set containing the variables age, gender and
race

agents <- data.frame(
 id = 1:population_size,
 x = sample(1:grid_size, population_size, replace = TRUE),
 y = sample(1:grid_size, population_size, replace = TRUE),
 gender =samplehiv_data$Gender,
 age = samplehiv_data$Age,
 race = samplehiv_data$Race,
 health_status = "susceptible" # default initialized attribute which is
set to susceptible initially
```

## Simulating the effects of HIV/AIDs on human population and exploring the management option using Agent Based Modelling and Integrating Machine learning for Predictive Disease Outcome Analysis

```
)

Using the HIV_diagnoses variable from the hiv dataset to determine
initial infected agents

infected_agents <- sample(1:population_size,
sum(samplehiv_data$HIV_diagnoses > 0), replace = FALSE)
agents$health_status[infected_agents] <- "infected"

...

#Exploring the base line scenario 1 to simulate the transmission dynamics
of HIV/AIDs disease without disease intervention

```{r}
# Setting the Simulation parameters for the simulation process

infection_probability <- 0.05 # Infection probability Infection rate per
interaction
progression_probability <- 0.10 # Probability to progress from HIV to AIDS
death_probability <- 0.05 # Probability to die from AIDS
simulation_steps <- 100 # Number of time steps which represent 100 days
for this simulation

# Storing the results in data frame

results <- data.frame(step = integer(), susceptible = integer(), infected =
integer(), aids = integer(), dead = integer())

# Running the simulation
for (step in 1:simulation_steps) {
  # Moving agents randomly on the grid
  agents$x <- pmax(pmin(agents$x + sample(-1:1, population_size, replace =
TRUE), grid_size), 1)
  agents$y <- pmax(pmin(agents$y + sample(-1:1, population_size, replace =
TRUE), grid_size), 1)

  # Spreading of HIV through interactions using the loop
  for (i in 1:population_size) {
    if (agents$health_status[i] == "infected") {
      neighbors <- which(agents$x == agents$x[i] & agents$y == agents$y[i])
      susceptible_neighbors <- neighbors[agents$health_status[neighbors] ==
"susceptible"]
      if (length(susceptible_neighbors) > 0) {
        infected <- runif(length(susceptible_neighbors)) <
infection_probability
        agents$health_status[susceptible_neighbors[infected]] <- "infected"
      }
    }
  }

  # Progression to AIDS based on AIDS_diagnoses variable from the dataset

  infected_indices <- which(agents$health_status == "infected")
  for (i in infected_indices) {
```

Simulating the effects of HIV/AIDs on human population and exploring the management option using Agent Based Modelling and Integrating Machine learning for Predictive Disease Outcome Analysis

```
aids_diagnosis_probability <-
samplehiv_data$AIDS_diagnoses[sample_indices[i]]

# Handling any NA values by assigning a default probability or skipping
if (is.na(aids_diagnosis_probability)) {
  aids_diagnosis_probability <- 0.05 # Default probability if any NA
  occurred
}

if (runif(1) < aids_diagnosis_probability) {
  agents$health_status[i] <- "aids"
}
}

# calculating the Deaths based on the disease progression

aids_indices <- which(agents$health_status == "aids")
deaths <- runif(length(aids_indices)) < death_probability
agents$health_status[aids_indices[deaths]] <- "dead"

# Recording the results
results <- rbind(results, data.frame(
  step = step,
  susceptible = sum(agents$health_status == "susceptible"),
  infected = sum(agents$health_status == "infected"),
  aids = sum(agents$health_status == "aids"),
  dead = sum(agents$health_status == "dead")
))

# the simulation output status for each step
cat("Step", step, ":", table(agents$health_status), "\n")
}

# Plotting the number of agents in different health states over time
ggplot(results, aes(x = step)) +
  geom_line(aes(y = susceptible, color = "Susceptible")) +
  geom_line(aes(y = infected, color = "Infected")) +
  geom_line(aes(y = aids, color = "AIDS")) +
  geom_line(aes(y = dead, color = "Dead")) +
  labs(y = "Number of Agents", color = "Health Status") +
  theme_minimal()

results_long <- melt(results, id.vars = "step", variable.name =
"health_status", value.name = "count")

ggplot(results_long, aes(x = step, y = count, fill = health_status)) +
  geom_area(alpha = 0.6) +
  labs(title = "Proportions of Health Status Over Time", x = "Time Step", y
= "Count") +
  theme_minimal()

...

#Exploring scenario 2 simulation of transmission of HIV/AIDS across
demographic factors stratified by age,gender ,and race
```

Simulating the effects of HIV/AIDs on human population and exploring the management option using Agent Based Modelling and Integrating Machine learning for Predictive Disease Outcome Analysis

```
`r`{r}

# Categorizing the age groups
agents$age_group <- agents$age
agents$race_group <- agents$race

# Storing the results stratified by demographic factors
results_by_demographics <- list()

# Running the simulation
for (step in 1:simulation_steps) {
  # Move agents randomly
  agents$x <- pmax(pmin(agents$x + sample(-1:1, population_size, replace =
TRUE), grid_size), 1)
  agents$y <- pmax(pmin(agents$y + sample(-1:1, population_size, replace =
TRUE), grid_size), 1)

  # Spreading HIV through interactions through neighbour agents
  for (i in 1:population_size) {
    if (agents$health_status[i] == "infected") {
      neighbors <- which(agents$x == agents$x[i] & agents$y == agents$y[i])
      susceptible_neighbors <- neighbors[agents$health_status[neighbors] ==
"susceptible"]
      if (length(susceptible_neighbors) > 0) {
        infected <- runif(length(susceptible_neighbors)) <
infection_probability
        agents$health_status[susceptible_neighbors[infected]] <- "infected"
      }
    }
  }

  # Progression to AIDS based on AIDS_diagnoses variable through the
interaction from neighbour agents
  infected_indices <- which(agents$health_status == "infected")
  for (i in infected_indices) {
    aids_diagnosis_probability <- samplehiv_data$AIDS_diagnoses[i]

    # Handling any NA values by assigning a default probability or skipping

    if (is.na(aids_diagnosis_probability)) {
      aids_diagnosis_probability <- 0.05 # Default probability if NA
values occurred
    }

    if (runif(1) < aids_diagnosis_probability) {
      agents$health_status[i] <- "aids"
    }
  }

  # Deaths
  aids_indices <- which(agents$health_status == "aids")
  deaths <- runif(length(aids_indices)) < death_probability
  agents$health_status[aids_indices[deaths]] <- "dead"

  # Recording the results stratified by demographic factors
  results_by_demographics[[step]] <- agents %>%
    group_by(age_group, race_group, gender, health_status) %>%
```

Simulating the effects of HIV/AIDs on human population and exploring the management option using Agent Based Modelling and Integrating Machine learning for Predictive Disease Outcome Analysis

```
    summarize(count = n(), .groups = 'drop')

# Output the simulation status for each step
cat("Step", step, ":", table(agents$health_status), "\n")
}

# Combining the results across all steps
results_demographics_combined <- bind_rows(lapply(results_by_demographics,
function(x) x %>% mutate(step = step)))

# Calculating the difference (change) in the number of infected agents over
time
results_demographics_combined <- results_demographics_combined %>%
  group_by(age_group, gender, health_status) %>%
  arrange(step) %>%
  mutate(diff_count = count - lag(count, default = first(count)))

# Plotting the number of infected agents by age group over time
ggplot(results_demographics_combined %>% filter(health_status ==
"infectd"),
  aes(x = step, y = count, color = age_group)) +
  geom_line() +
  labs(title = "Infected Agents by Age Group Over Time", x = "Time Step", y
= "Number of Infected Agents") +
  theme_minimal()

# Plotting the number of infected agents by race group over time
ggplot(results_demographics_combined %>% filter(health_status ==
"infectd"),
  aes(x = step, y = count, color = race_group)) +
  geom_line() +
  labs(title = "Infected Agents by Race Group Over Time", x = "Time Step",
y = "Number of Infected Agents") +
  theme_minimal()

# Comparing the gender differences in progression to AIDS
ggplot(results_demographics_combined %>% filter(health_status == "aids"),
  aes(x = step, y = count, color = gender)) +
  geom_line() +
  labs(title = "Progression to AIDS by Gender Over Time", x = "Time Step",
y = "Number of Agents with AIDS") +
  theme_minimal()
```

#Exploring scenario analysis 3 for intervention strategies Viral
suppression and linked to care impact on the HIV/AIDS transmission

```{r}

# Sampling data from the hiv dataset to assign initial health status
sample_indices <- sample(1:nrow(hiv_df), population_size, replace = TRUE)
sample_data <- hiv_df[sample_indices, ]

# Replacing if any missing values in the viral suppression and linked to
care columns with reasonable defaults
sample_data$Viral_suppression[is.na(sample_data$Viral_suppression)] <- 0
```

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```
sample_data$Linked_to_care_3_months[is.na(sample_data$Linked_to_care_3_months)] <- 0

# Initializing agents attribute with variables viral suppression and
linked to care

agents <- data.frame(
  id = 1:population_size,
  x = sample(1:grid_size, population_size, replace = TRUE),
  y = sample(1:grid_size, population_size, replace = TRUE),
  gender = sample_data$Gender,
  age = sample_data$Age,
  race = sample_data$Race,
  health_status = "susceptible",
  viral_suppressed = runif(population_size) <
rep(sample_data$Viral_suppression, length.out = population_size),
  linked_to_care = runif(population_size) <
rep(sample_data$Linked_to_care_3_months, length.out = population_size),
  aids_diagnosis = sample_data$AIDS_diagnoses > 0 # AIDS diagnosis status
)

# Using the HIV_diagnoses to determine initial infected agents
infected_agents <- sample(1:population_size, sum(sample_data$HIV_diagnoses
> 0), replace = FALSE)
agents$health_status[infected_agents] <- "infected"

# Simulating loop with intervention strategy

results <- data.frame(
  step = integer(),
  susceptible = integer(),
  infected = integer(),
  aids = integer(),
  dead = integer(),
  viral_suppressed = integer(),
  linked_to_care = integer()
)

for (step in 1:simulation_steps) {
  # Increasing viral suppression efforts over time, especially for those
agents with linked to care
  suppression_increase <- 0.01 * step / simulation_steps # Linear increase
in viral suppression efforts

  for (i in 1:population_size) {
    if (!is.na(agents$linked_to_care[i]) && agents$linked_to_care[i]) {
      agents$viral_suppressed[i] <- agents$viral_suppressed[i] | (runif(1)
< (suppression_increase * 1.5)) # Setting Higher
suppression increase rate if linked to care
    } else {
      agents$viral_suppressed[i] <- agents$viral_suppressed[i] | (runif(1)
< suppression_increase) # Normal suppression
increase rate
    }
  }

  # Moving agents randomly on the grid
```

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```
agents$x <- pmax(pmin(agents$x + sample(-1:1, population_size, replace =
TRUE), grid_size), 1)
agents$y <- pmax(pmin(agents$y + sample(-1:1, population_size, replace =
TRUE), grid_size), 1)

# Spreading HIV through interactions by non-suppressed infected agents

for (i in 1:population_size) {
  if (agents$health_status[i] == "infected" &&
!agents$viral_suppressed[i]) {
    neighbors <- which(agents$x == agents$x[i] & agents$y == agents$y[i])
    susceptible_neighbors <- neighbors[agents$health_status[neighbors] ==
"susceptible"]
    if (length(susceptible_neighbors) > 0) {
      infected <- runif(length(susceptible_neighbors)) <
infection_probability
      agents$health_status[susceptible_neighbors[infected]] <- "infected"
    }
  }
}

# Progression to AIDS less likely if agents are viral suppressed,
linked to care, and no AIDS diagnosis
infected_indices <- which(agents$health_status == "infected")
if (length(infected_indices) > 0) {
  non_suppressed_infected <-
infected_indices[!agents$viral_suppressed[infected_indices]]
  suppressed_infected <-
infected_indices[agents$viral_suppressed[infected_indices]]

  if (length(non_suppressed_infected) > 0) {
    progress_to_aids_non_suppressed <-
runif(length(non_suppressed_infected)) < (progression_probability +
agents$aids_diagnosis[non_suppressed_infected] * 0.1)

agents$health_status[non_suppressed_infected[progress_to_aids_non_suppressed]] <- "aids"
  }

  if (length(suppressed_infected) > 0) {
    progress_to_aids_suppressed <- runif(length(suppressed_infected)) <
(progression_probability * 0.5 + agents$aids_diagnosis[suppressed_infected]
* 0.05)

agents$health_status[suppressed_infected[progress_to_aids_suppressed]] <-
"aids"
  }

# Reducing further progression to AIDS for those agents who are linked
to care

if (length(infected_indices) > 0) {
  linked_to_care_infected <-
infected_indices[agents$linked_to_care[infected_indices]]
  progress_to_aids_linked_to_care <-
runif(length(linked_to_care_infected)) < (progression_probability * 0.5 +
agents$aids_diagnosis[linked_to_care_infected] * 0.05)
```

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```
agents$health_status[linked_to_care_infected[progress_to_aids_linked_to_care]] <- "aids"
  }
}

# Deaths less likely if virally suppressed or linked to care

aids_indices <- which(agents$health_status == "aids")
if (length(aids_indices) > 0) {
  non_suppressed_aids <-
aids_indices[!agents$viral_suppressed[aids_indices]]
  suppressed_aids <- aids_indices[agents$viral_suppressed[aids_indices]]

  if (length(non_suppressed_aids) > 0) {
    deaths_non_suppressed <- runif(length(non_suppressed_aids)) <
death_probability
    agents$health_status[non_suppressed_aids[deaths_non_suppressed]] <-
"dead"
  }

  if (length(suppressed_aids) > 0) {
    deaths_suppressed <- runif(length(suppressed_aids)) <
(death_probability * 0.5)
    agents$health_status[suppressed_aids[deaths_suppressed]] <- "dead"
  }

  # Further reducing the death probability for those agents who are
linked to care
  if (length(aids_indices) > 0) {
    linked_to_care_aids <-
aids_indices[agents$linked_to_care[aids_indices]]
    deaths_linked_to_care <- runif(length(linked_to_care_aids)) <
(death_probability * 0.5)
    agents$health_status[linked_to_care_aids[deaths_linked_to_care]] <-
"dead"
  }
}

# Recording the results
results <- rbind(results, data.frame(
  step = step,
  susceptible = sum(agents$health_status == "susceptible"),
  infected = sum(agents$health_status == "infected"),
  aids = sum(agents$health_status == "aids"),
  dead = sum(agents$health_status == "dead"),
  viral_suppressed = sum(agents$viral_suppressed, na.rm = TRUE),
  linked_to_care = sum(agents$linked_to_care, na.rm = TRUE)
))

cat("Step", step, ":", table(agents$health_status), "\n")
}

# Plotting the number of agents in different health states over time
ggplot(results, aes(x = step)) +
  geom_line(aes(y = susceptible, color = "Susceptible")) +
  geom_line(aes(y = infected, color = "Infected")) +
  geom_line(aes(y = aids, color = "AIDS")) +
```


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```
geom_line(aes(y = dead, color = "Dead")) +
geom_line(aes(y = viral_suppressed, color = "Viral Suppressed")) +
geom_line(aes(y = linked_to_care, color = "Linked to Care")) +
labs(y = "Number of Agents", color = "Health Status") +
theme_minimal()

# Transforming the results for area plot
results_long <- melt(results, id.vars = "step", variable.name =
"health_status", value.name = "count")

# Plotting proportions of health status over time
ggplot(results_long, aes(x = step, y = count, fill = health_status)) +
  geom_area(alpha = 0.6) +
  labs(title = "Proportions of Health Status Over Time", x = "Time Step", y
= "Count") +
  theme_minimal()

...

#.....Machine learning Algorithm....
.....

#Applying Random Forest Regression algorithm on the output generated by the
scenario 3 simulation to predict future disease outcomes.

```{r}
Feature Engineering process: Create lagged features on the health status
results output generated by the agent based model to predict the next
step's 'infected' count to capture temporal dependencies

results <- results %>% mutate(
 susceptible_lag1 = lag(susceptible, 1),
 infected_lag1 = lag(infected, 1),
 aids_lag1 = lag(aids, 1),
 dead_lag1 = lag(dead, 1)

) %>% na.omit() # Removing any rows with NA values due to lagging

set random seed
set.seed(123) # Set seed for reproducibility

Splitting the data into training and testing sets
trainIndex <- createDataPartition(results$infected, p = .7, list = FALSE)
trainData <- results[trainIndex,]
testData <- results[-trainIndex,]

Training a Random Forest model to predict the number of infected
individuals
set.seed(123)

#Defining a formula for predicting the future infected individual counts
results_formula <- infected ~ susceptible_lag1 + aids_lag1 + dead_lag1 +
viral_suppressed + linked_to_care + step

rf_model <- randomForest(results_formula,
 data = trainData,
 importance = TRUE,
 ntree = 500)
```

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```
Printing model summary
print(rf_model)

Plotting the Variable importance plot
varImpPlot(rf_model)

Predicting on the test set
rf_predictions <- predict(rf_model, testData)

Evaluating the performance of the model
MAE <- mean(abs(rf_predictions - testData$infected))
cat("Mean Absolute Error (MAE):", MAE, "\n")

Plotting the predicted vs actual values
ggplot() +
 geom_point(aes(x = testData$step, y = testData$infected, color =
"Actual")) +
 geom_line(aes(x = testData$step, y = rf_predictions, color =
"Predicted"), size = 1) +
 labs(title = "Predicted vs Actual Infected Counts (Random Forest)", x =
"Time Step", y = "Number of Infected") +
 theme_minimal() +
 scale_color_manual(name = "Legend", values = c("Actual" = "blue",
"Predicted" = "red"))

```
```